



Affirmed Networks Troubleshooting Guide

Date: January 2022

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A Microsoft Company

Affirmed Networks Troubleshooting Guide
Revision: A

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Preface

This guide provides information to help identify and troubleshoot common issues associated with Affirmed Networks software.

Note: *The Affirmed Networks Mobile Content Cloud™ (MCC) is the mobile services software solution that operates on virtual platforms such as blade systems.*

Audience

This guide is intended for network operators (sometimes referred to as Users or Operators in this guide), installation staff, and other qualified service personnel responsible for installing, provisioning, and monitoring the system. The audience should be familiar with Long Term Evolution (LTE), 3G or 4G networks.

How to Use This Guide

This guide contains the following chapters:

Chapter/Appendix	Describes
Diagnosing Issues and Performing Health Checks	Procedures to check the system health and diagnose issues using the CLI.
Session Management and Debugging	Troubleshooting problems related to bearer activation and bearer traffic, and recovering from a fatal error.
Configuring Interfaces	Information on configuring interfaces.
Technical Support	Accessing the Affirmed Networks support ticket system and escalation contacts for customer support.

Conventions

This guide uses the following conventions, when applicable:

Description	Convention	Examples
Command names, keywords, dialog boxes, buttons, icons, menu	Times New Roman bold font	Use configure_ha to configure... Enter y . To continue, press Enter .
Selecting a menu item	Menu => Option	Select Security => Group Management
Variable parameters for which you supply values	<courier bold italic blue font>	configure_dr --host <host name>
User input	Courier bold blue font	cd /opt/Affirmed/NMS/bin
Screen messages, system output, sample source code	Courier regular blue font	Checking for License...
Required keywords and arguments in square brackets	[]	cluster [id] node [id] reboot
Optional keywords and arguments in curly brackets	{ }	{cdr-file-name}
Keywords separated by the pipe symbol. Requires you to specify only one item from the set.		Is this correct? [y n]
Keywords representing a single element in angle brackets	< >	<host name>
Asterisk after a CLI object indicates a required object	*	name*
Variables, book and chapter titles, file path names, new terms, and emphasized text	<i>Times New Roman italic font</i>	See the <i>Affirmed Networks Troubleshooting Guide</i> . <i>Workflow</i> is an application running on the network element.
Images in online help		Many images can be enlarged by clicking the image with the left-mouse button. Alternately, they can be enlarged/reduced by pressing the Enter key when the image is highlighted (surrounded by a thick border) on the screen.

Note: Additional information that may apply to the subject text.

Tip: Helpful suggestions.

Caution: Proceed carefully to avoid possible equipment damage or data loss.

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See the *Affirmed Networks Documentation Library* for a complete list of Affirmed Networks product documentation.

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Diagnosing Issues and Performing Health Checks

This section describes the procedures to check the system health and diagnose issues using the Command Line Interface (CLI).

Displaying Command Line Interface Command Output

CLI command output displays either horizontally or vertically, depending on the width of the terminal window. Both orientations are used in this guide. For example:

Horizontal output:

```
an3000-mcm-slot7cpu1# show extended-storage status

extended-storage status

      ADMIN  OPER      CURRENT  CURRENT
STATE   ID  SLOT  STATE  STATUS  DISK PARTITION  OPERATION  STATE
-----
active  28   8    unused down    extended-storage none       none
standby 28   7    unused down    extended-storage none       none
```

Vertical output:

```
extended-storage status active 28 8
admin-state           unused
oper-status           down
disk-partition        extended-storage
current-operation     none
current-state         none
extended-storage status standby 28 7
admin-state           unused
oper-status           down
disk-partition        extended-storage
current-operation     none
current-state         none
```

You can copy all CLI command input and output and save it to an external file. This information may be used by Affirmed Networks Support to diagnose issues.

Subscriber Commands

Use the subscriber commands to display real-time information about subscriber data.

show subscriber summary

Use the **show subscriber summary** command to display information about individual subscribers by IMSI, APN, and other criteria.

*Note: Do not run the **show subscriber summary** command without including additional parameters. This command displays **all** subscribers and is resource intensive.*

Use the following parameters to display more information about attached subscribers:

Table 1-1 CLI Subscriber Command Parameters

Parameter	Description
active-only	Only select active subscribers
apn	Specific APN value (case insensitive)
bearer-type	Only select subscribers with a specific bearer/PDP type
control-plane-session-only	Only select control plane sessions
credited-only	Only select credited subscribers
cups-peer	Specific CUPS(Control or User Plane) Peer's name
cups-profile-id	CUPS(Control or User Plane) Profile Id
cups-session-id	Specific CUPS(Control or User Plane) Session Id Value
database-type	Only select subscribers with a specific database type
datapfile-name < data_profile_name >	Select subscribers that are using, or have applied, the specified workflow data-profile name
default-gx-policy	Select subscribers for PGW sessions that use a Default Gx Policy based on PCRF connection error handling. Applicable to PGW sessions only. Not applicable to SGW/user plane sessions (including MEC deployments)
emergency-only	Only select emergency subscribers
end-imsi	End IMSI value, inclusive
exclude-indirect-forwarding-tunnels	Exclude indirect forwarding tunnels

Parameter	Description
fsapi	Only select subscribers with a specific FSAPI value
gateway-name	Gateway name
gateway-type	Gateway type
gx-failure	Select subscribers for sessions that fail to interact with the PCRF due to a Gx interface failure. Includes sessions for which Default Gx Policy is enabled and sessions using Gx Rejoin that are waiting to re-connect to a PCRF. Applicable to PGW sessions, and to SGW sessions for MEC deployments
gy-failure-detected	Select subscribers for which the PCRF has armed the Cisco credit-control trigger for gy-failure and gy-credit-control failure was detected for one or more online rules installed or activated by PCRF. Applicable to PGW session only
home-only	Only select home subscribers
idle-only	Only select idle subscribers
imsi	Specific IMSI value
indirect-forwarding-tunnels-only	Only select indirect forwarding tunnels
ip-address	Only select subscribers allocated a specific IP Address
ip-address-allocated	Only select subscribers who have IP address allocated
mac-address	MAC address
msisdn	Specific MSISDN value
rat-type	Only select subscribers using specific Radio Access Technology
service-name < <i>gateway_task_name</i> >	Select subscribers for sessions that are anchored to the specified gateway task's service name
service-rules-greater-than < <i>integer_value</i> >	Select subscribers installed with a number of service rules greater than or equal to the specified value. Applicable to PGW sessions only. Not applicable to SGW/user plane sessions
standalone [true false]	If true, select subscribers for standalone sessions only. If false, select subscribers for co-located sessions only. Applicable to PGW, SGW, and GwAgent sessions. Can be combined with existing filters (for example, gateway-type pgw-c/pgw to select PGW CUPS or non-CUPS standalone sessions)
start-imsi	Start IMSI value, inclusive
subscriber-nai	Subscriber NAI
uptime-greater-than	Only select subscribers up for longer than specified number of seconds

Parameter	Description
uptime-less-than	Only select subscribers up for less than specified number of seconds
using-default-quota	Only select subscriber using default quota
visiting-only	Only select visiting subscribers

For example:

```
an3000-mcm-slot7cpu1# show subscriber summary imsi 226041000000003
```

```
subscriber show-summary 98304
```

```

imsi                226041000000003
gateway-name        UPLANE-1
gateway-type        saegw-u
meid                999900000000030
msisdn              40010101700002
apn                 sfr.consumer
service-name        GwAgent_6_9_1_2
subscriber-activity-state n/a
subscriber-location-state n/a
subscriber-nai       ""
uptime-in-seconds    13464
mac-address          ""
cups-session-id      294912
cups-peer            SXB-V4-PEER1
sub-memory-storage-state in-memory
default-bearer-id    5

```

```
an3000-mcm-slot7cpu1# show subscriber summary rat-type eutran
```

```

imsi                226041000000003
gateway-name        NYsgw-1
gateway-type        sgw-c
meid                999900000000030
msisdn              40010101700002

```

```

apn                    sfr.consumer
service-name           SGW_6_9_1_7
subscriber-activity-state active
subscriber-location-state visiting
subscriber-nai         ""
uptime-in-seconds      13353
mac-address           ""
cups-session-id        98304
cups-peer              CL6-UPS-V4-PEER
sub-memory-storage-state in-memory
default-bearer-id      5

```

show subscriber pdn-session

Use the **show subscriber pdn-session** command to display information about a specific SGW or PGW session. This command is useful for troubleshooting data flows, because the PGW is the IP anchor point.

The **pdn-session** parameter displays a breakdown of a subscriber's session and provides important troubleshooting information

Note: To view the **subscriber-type** field in CUPS deployments running MCC release 9.3 or later, you must issue this command on the C-Plane. This field was removed from the U-Plane output.

Some of the more commonly used fields for troubleshooting a subscriber's PDN session are highlighted in the following example:

```

an3000-mcm-slot8cpu1# show subscriber pdn-session 262144

subscriber pdn-session 262144
default-bearer-id                    5
default-logical-bearer-id            0
ue-v4-ip-address                     22.114.1.2
peer-interface-type                  s5
number-of-bearers                    1
apn-name                            att.consumer
original-apn-name                    att.consumer
prior-apn-name                       att.consumer
incoming-apn-name                    att.consumer
reporting-apn-name                   att.consumer
access-in-control-teid-interface-type s5s8-pgw-gtpc

```

```

access-in-control-teid-network-ctxt      2
access-in-control-teid-ip-address-type   ipv4
access-in-control-teid-v4-ip-address     10.64.133.32
access-in-control-teid-value             262144
access-out-control-teid-interface-type    s5s8-sgw-gtpc
access-out-control-teid-ip-address-type   ipv4
access-out-control-teid-v4-ip-address     10.64.130.32
access-out-control-teid-value             239774
imsi                                      3571234578932
gateway-name                             pgw
gateway-type                             pgw
meid                                      998654325958932
msisdn                                    998654325958832
ULI-CGI                                  ""
ULI-SAI                                  ""
ULI-RAI                                  ""
ULI-TAI                                  "MCC:208 MNC:10"
TAC:291"
  ULI-ECGI                              "MCC:208 MNC:10"
ECI:74565"
  ULI-LAI                                ""
  UCI-CSG-ID                             36518
  UCI-CSG-Access-Mode                     2
  UCI-CSG-CMI                             0
  serving-network                         "MCC:208 MNC:10"
  pcscf-reselection-supported-by-ue        false
  pcscf-restoration-occured                false
  no. of p-cscf: 0
bearers: { pdn-session-id bearer-id }    262144 5
  is-cups-session                          false
service-data-flow-ids: { pdn-session-id service-type service-flow-id
service-rule-id }
  262144 proxy 2 2
nat info: { nat ip[allocation-policy] port-chunk-list }
  active-connections                       0
  total-connections                        28
  rejected-connections                     0
  workflow-data-profile                    dataProfile1
  workflow-control-profile                 wfctrlProf
  workflow-override-profile                "not set"
  workflow-location-index                  11
  workflow-active-chassis                  32
  workflow-active-slot                     4
  workflow-active-cpu                      1
  workflow-gn-next-hop-network-context     2
  workflow-gn-next-hop-ip                 10.64.32.1
  workflow-gn-next-hop-interface-index     1

```

uplink-ambr	10000
downlink-ambr	10000
uplink-burst-size	1250
downlink-burst-size	1250
inactivity-timer-duration	900
idle-timer-state	disabled
maximal-duration-timer-state	disabled
subscriber-type	4g
activation-time	2020-01-21T22:21:16-0500
pdn-session-duration	"1 days 15 hrs 0 mins 17 secs"
pdn-session-idle-duration	"1 days 14 hrs 44 mins 13 secs"
pdn-session-monitor-timer-state	not-running
received-pfcp-message-in-monitor-interval	false
pdn-session-state	idle
uplink-green-packets	265
uplink-green-bytes	72453
uplink-yellow-packets	0
uplink-yellow-bytes	0
uplink-red-packets	0
uplink-red-bytes	0
uplink-tcp-retransmission-packets	0
uplink-tcp-retransmission-bytes	0
uplink-diverted-packets	245
uplink-error-drop-packets	7
uplink-error-drop-bytes	1004
uplink-service-drop-packets	0
uplink-service-drop-bytes	0
uplink-no-tft-packets	0
uplink-no-route-packets	0
uplink-ttl-exhaust-packets	7
uplink-bad-header-packets	0
uplink-bad-ip-addr-packets	0
uplink-mtu-discard-packets	0
uplink-transmit-icmp-packets	0
uplink-send-error-packets	0
uplink-send-error-bytes	0
uplink-router-solicitation-packets	0
Uplink-router-solicitation-bytes	0
uplink-neighbour-solicitation-packets	0
uplink-neighbour-solicitation-bytes	0
downlink-green-packets	285
downlink-green-bytes	134552
downlink-yellow-packets	0
downlink-yellow-bytes	0

downlink-red-packets	0
downlink-red-bytes	0
downlink-tcp-retransmission-packets	0
downlink-tcp-retransmission-bytes	0
downlink-diverted-returned-packets	265
downlink-error-drop-packets	0
downlink-error-drop-bytes	0
downlink-service-drop-packets	0
downlink-service-drop-bytes	0
downlink-no-tft-packets	0
downlink-no-route-packets	0
downlink-ttl-exhaust-packets	0
downlink-mtu-discard-packets	0
downlink-transmit-icmp-packets	4
downlink-bad-header-packets	0
downlink-send-error-packets	0
downlink-send-error-bytes	0
downlink-router-advertisement-packets	0
downlink-router-advertisement-bytes	0
downlink-neighbour-advertisement-packets	0
downlink-neighbour-advertisement-bytes	0
uplink-hairpin-policy-discard-packets	0
uplink-hairpin-error-discard-packets	0
uplink-tunneled-packets	0
uplink-aggregate-bw-pkts-dropped	0
uplink-aggregate-bw-bytes-dropped	0
downlink-aggregate-bw-pkts-dropped	0
downlink-aggregate-bw-bytes-dropped	0
uplink-appgrp-aggregate-bw-pkts-dropped	0
uplink-appgrp-aggregate-bw-bytes-dropped	0
downlink-appgrp-aggregate-bw-pkts-dropped	0
downlink-appgrp-aggregate-bw-bytes-dropped	0
uplink-flowauth-pkts-dropped	0
uplink-flowauth-bytes-dropped	0
downlink-flowauth-pkts-dropped	0
downlink-flowauth-bytes-dropped	0
security-did-not-forward-pkts-dropped	0
rat-type	E-UTRAN
pdn-type	ipv4
workflow-subscriber-analyzer	Analyze1
workflow-data-profile-sub-analyzer	Analyze1
l2tp-tunnel-endpoint	::
l2tp-tunnel-id	0
l2tp-session-id	0
l2tp-local-tunnel-id	0
l2tp-local-session-id	0
l2tp-tunnel-assignment-id	""

delay-tolerant-pdn-connection-enabled	false
low-priority-access-indicated-pdn	false
control-plane-only-pdn-connection	false
suspended	false
guard-timer-running	false
time-zone	"GMT -21:15 hours
(0xff) "	
emergency-session	false
user-identifier-type	imsi
tethering	false
origination-timestamp	0
max-wait-time	0
access-point-mac-address	" "
total-tethered-connections	0
active-tethered-connections	0
pcscf-table-name	N/A
pco-ipv4-link-mtu-size	0
pco-nonip-link-mtu-size	0
pco-pcscf-fqdn-used-for-dns-query	" "
uplink-splmn-pkt-rate-valid	false
downlink-splmn-pkt-rate-valid	false
apn-pkt-rate-timeunit	not-set
apn-pkt-rate-exception-report	not-set
uplink-apn-pkt-rate-valid	false
cc-group	" "
pco-action	0
is-using-default-quota	false
crbn-profiles	" "
charging-rule-names	" "
aggregate-qos-policy-id	0
uplink-peak-throughput	0
downlink-peak-throughput	0
uplink-activity-duration	0
downlink-activity-duration	0
session-terminate-pkts-dropped	0
proxy-bypassed-connections	0
session-terminate-bytes-dropped	0
tethered-flows-detected-using-ttl	0
tethered-flows-detected-using-user-agent	0
tethering-entitled	false
tethering-entitled-flows	0
content-filter-instance-name	" "
tethered-flows-detected-using-tcp-ip-fingerprinting	0
tethered-flows-detected-using-gquic-user-agent	0
pcrf-update-down-avoid-pcrf	false
transparent-data	" "
user-plane-dns-fqdn	" "

```
charging-characteristics-value 0
master-crbn-profile ""
tethered-flows-reported-using-ttl 0
tethered-flows-reported-using-user-agent 0
tethered-flows-reported-using-tcp-ip-fingerprinting 0
tethered-flows-reported-using-gquic-user-agent 0
total-session-qos-policing-drops 0
```

show subscriber pdn-service-data-flow

When troubleshooting issues specifically related to QoS, proxy, and charging, use parameters that control a particular subscriber flow.

The output from the `show subscriber pdn-session` command (highlighted in blue above) displays the flow details:

```
bearers: { pdn-session-id bearer-id }

262144 5

is-cups-session true
is-pfcp-std-mode false

service-data-flow-ids: { pdn-session-id service-type
service-flow-id service-rule-id }

262144 dns-proxy 2 12

262144 qos 2 42
```

This information is used to determine whether the user is, for example, using the correct service-flow-name. This command also displays specific information about workflow and service data types. The subscriber trace does not include public post-NAT UE IP address information.

Note: This command only displays the current status of the subscriber information, which is from the SGW/PGW side pre-Proxy. No NAT mapping information is available in the current trace feature.

Use the `show subscriber pdn-service-data-flow` command and parameters to display specific individual service data flows. For example:

```
an3000-mcm-slot7cpu1# show subscriber pdn-service-data-flow 262144
```

```
Description: PDN Service Data Flow Level Information (key:
<pdn-session-id> <service-type> <service-flow-id> <service-
rule-id>)
```

Possible completions:

```
accounting          charging          content-cache
content-filter      dhcp             dns-proxy
```

latency	mdns	metering
mirror	nat	proxy
qos	sip-proxy	smtp-proxy
steering	unknown	video-adaptation
wap-gateway		<cr>

```
an3000-mcm-slot8cpu1# show subscriber pdn-service-data-flow
262144 qos 2 42
```

```
subscriber pdn-service-data-flow 262144 qos 2 42
```

service-flow-name	QOS-FLOW-PARG
service-rule-name	SR-PARG
imsi	2260410000000020
gateway-name	NYpgw-1
gateway-type	pgw
meid	9999000000000320
msisdn	9999000000000320
correlation-id	0
service-data-flow-id	35
total-connections	3512
total-packets	128550
total-drop-packets	0

show subscriber count

Use the **show subscriber count** command to display the global number of subscribers.

Use the parameters in [CLI Subscriber Command Parameters](#) to define a specific category or range of subscribers.

For example:

```
an3000-mcm-slot7cpu1# show subscriber count <category/range>
```

subscriber clear

In certain situations, you may need to clear user(s) from the MCC. Use the **subscriber clear** command and the parameters in [CLI Subscriber Command Parameters](#).

For example:

```
an3000-mcm-slot7cpu1# subscriber clear imsi 123456789876543
```

subscriber clear-local

Use the **subscriber clear-local** command to clear all matching subscribers locally without informing peers. This will not clear subscribers associated with a standby geo-redundancy group.

Use the parameters in [CLI Subscriber Command Parameters](#) to define a specific category or range of subscribers.

Cluster Commands

Use the cluster commands to diagnose issues and display statistics.

The primary OAM interface for each cluster is through the management node (the MCM), which is implemented on nodes (VMs) 7 and 8 on each cluster.

show cluster summary

Use the `show cluster summary` command to display information about each VM, including admin state, HA state, and uptime.

An asterisk at the end of the release number (highlighted in green below) indicates that one or more tasks on the node is not yet operationally up. Depending on the configuration, some tasks can take longer than others to come up. When all nodes are operationally up, the cluster is fully operational.

A CPU state of down (highlighted in yellow below) indicates a more general fault on the VM or blade which could mean there is a problem with the underlying hypervisor layer (VMware) or a physical issue on the blade.

The asterisks are also seen during a VM reboot as processes startup, but are mostly seen during genuine fault scenarios (or conditions).

***Note:** If all the VMs on one physical blade are down, the problem is likely to be on the hypervisor layer or the physical blade.*

```
an3000-mcm-slot7cpu1# show cluster summary

cluster 32
summary 1 1
  Type          v-csm
  Personality    CPM
  Model          vCSM
  Admin State    enabled
  HA State       active
  CPU State      up
  CPU Uptime     "D:005 H:03 M:37 S:13"
  Release        12.0.0.0-1115.REL
summary 2 1
  Type          v-csm
  Personality    CPM
  Model          vCSM
  Admin State    enabled
  HA State       active
  CPU State      up
  CPU Uptime     "D:005 H:03 M:37 S:20"
  Release        12.0.0.0-1115.REL*
```

```
summary 3 1
Type          v-ssm
Personality    ISM
Model          vSSM
Admin State    enabled
HA State       active
CPU State      down
CPU Uptime     "D:005 H:03 M:37 S:28"
Release        12.0.0.0-1115.REL
```

show cluster node task

Use the **show cluster node task** command to display the status of the task running under each node and its associated service. For example:

```
an3000-mcm-slot7cpu1# show cluster node task
```

```
cluster 125
node 1
task dryup
```

CPU			HA	OPER		DRYUP
NUMBER	TASK NAME	SERVICE NAME	STATE	STATE	DRYUP STATUS	LOAD

cpu1	EdrTask_125_1_1_EdrTask_0	EdrTask_125_1_1	active	up	Not applicable	0
cpu1	EventConfigRelayAgent_125_1_1_EventConfigRelayAgent_1	EventConfigRelayAgent_125_1_1	active	up	Not applicable	0
cpu1	FMAgent_125_1_1_FMAgent_1	FMAgent_125_1_1	active	up	Not applicable	0
cpu1	FPexecMaint_125_1_1_FPexecMaint_0	FPexecMaint_125_1_1	active	up	Not applicable	0
cpu1	FileAgent_125_1_1_FileAgent_1	FileAgent_125_1_1	active	up	Not applicable	0
cpu1	FpUeIpHdlr_125_1_1_0_FpUeIpHdlr_0	FpUeIpHdlr_125_1_1_0	active	up	Not applicable	0
cpu1	FpUeIpHdlr_125_1_1_1_FpUeIpHdlr_1	FpUeIpHdlr_125_1_1_1	active	up	Not applicable	0
cpu1	GwStatsAgent_125_1_1_GwStatsAgent_1	GwStatsAgent_125_1_1	active	up	Not applicable	0
cpu1	HwAgent_125_1_1_HwAgent_2	HwAgent_125_1_1	active	up	Not applicable	0
cpu1	ImAgent_125_1_1_ImAgent_1	ImAgent_125_1_1	active	up	Not applicable	0
cpu1	NRA_125_1_1_NRA_1	NRA_125_1_1	active	up	Not applicable	0
cpu1	NSAgent_125_1_1_NSAgent_0	NSAgent_125_1_1	active	up	Not applicable	0
cpu1	NmdAgent_125_1_1_NmdAgent_1	NmdAgent_125_1_1	active	up	Not applicable	0
cpu1	SecPolicyAgt_125_1_1_SecPolicyAgt_0	SecPolicyAgt_125_1_1	active	up	Not applicable	0
cpu1	ShadowTask_ShadowTask_0	ShadowTask	active	up	Not applicable	0
cpu1	TosfpFlowAgt_125_1_1_TosfpFlowAgt_0	TosfpFlowAgt_125_1_1	active	up	Not applicable	0
cpu1	TosfpGwHdlr_125_1_1_0_TosfpGwHdlr_0	TosfpGwHdlr_125_1_1_0	active	up	Not applicable	0
cpu1	TosfpGwHdlr_125_1_1_1_TosfpGwHdlr_1	TosfpGwHdlr_125_1_1_1	active	up	Not applicable	0
cpu1	TosfpTblDist_TosfpTblDist_1	TosfpTblDist	standby	up	Not applicable	0
cpu1	TransProxyAgt_125_1_1_TransProxyAgt_0	TransProxyAgt_125_1_1	active	up	Not applicable	0
cpu1	TransProxySipAlgAgt_125_1_1_TransProxySipAlgAgt_0	TransProxySipAlgAgt_125_1_1	active	up	Not applicable	0

```
task operation-detail
```

CPU			HA	OPER	
NUMBER	TASK NAME	SERVICE NAME	STATE	STATE	OPER REASON

cpu1	EdrTask_125_1_1_EdrTask_0	EdrTask_125_1_1	active	up	Not applicable
cpu1	EventConfigRelayAgent_125_1_1_EventConfigRelayAgent_1	EventConfigRelayAgent_125_1_1	active	up	Not applicable
cpu1	FMAgent_125_1_1_FMAgent_1	FMAgent_125_1_1	active	up	Not applicable
cpu1	FPexecMaint_125_1_1_FPexecMaint_0	FPexecMaint_125_1_1	active	up	Not applicable
cpu1	FileAgent_125_1_1_FileAgent_1	FileAgent_125_1_1	active	up	Not applicable
cpu1	FpUeIpHdlr_125_1_1_0_FpUeIpHdlr_0	FpUeIpHdlr_125_1_1_0	active	up	Not applicable
cpu1	FpUeIpHdlr_125_1_1_1_FpUeIpHdlr_1	FpUeIpHdlr_125_1_1_1	active	up	Not applicable
cpu1	GwStatsAgent_125_1_1_GwStatsAgent_1	GwStatsAgent_125_1_1	active	up	Not applicable
cpu1	HwAgent_125_1_1_HwAgent_2	HwAgent_125_1_1	active	up	Not applicable
cpu1	ImAgent_125_1_1_ImAgent_1	ImAgent_125_1_1	active	up	Not applicable
cpu1	NRA_125_1_1_NRA_1	NRA_125_1_1	active	up	All conditions asserted
cpu1	NSAgent_125_1_1_NSAgent_0	NSAgent_125_1_1	active	up	Not applicable
cpu1	NmdAgent_125_1_1_NmdAgent_1	NmdAgent_125_1_1	active	up	Not applicable
cpu1	SecPolicyAgt_125_1_1_SecPolicyAgt_0	SecPolicyAgt_125_1_1	active	up	Not applicable
cpu1	ShadowTask_ShadowTask_0	ShadowTask	active	up	Not applicable
cpu1	TosfpFlowAgt_125_1_1_TosfpFlowAgt_0	TosfpFlowAgt_125_1_1	active	up	Not applicable
cpu1	TosfpGwHdlr_125_1_1_0_TosfpGwHdlr_0	TosfpGwHdlr_125_1_1_0	active	up	Not applicable
cpu1	TosfpGwHdlr_125_1_1_1_TosfpGwHdlr_1	TosfpGwHdlr_125_1_1_1	active	up	Not applicable
cpu1	TosfpTblDist_TosfpTblDist_1	TosfpTblDist	standby	up	Not applicable
cpu1	TransProxyAgt_125_1_1_TransProxyAgt_0	TransProxyAgt_125_1_1	active	up	Not applicable
cpu1	TransProxySipAlgAgt_125_1_1_TransProxySipAlgAgt_0	TransProxySipAlgAgt_125_1_1	active	up	Not applicable

ask statistics

CPU		CPU	MEMORY	TASK
NUMBER	TASK NAME	UTILIZATION	UTILIZATION	UTILIZATION
cpu1	EdrTask_125_1_1_EdrTask_0	6.431	1.411	1.6
cpu1	EventConfigRelayAgent_125_1_1_EventConfigRelayAgent_1	0.1	0.18	1.6
cpu1	FMAgent_125_1_1_FMAgent_1	0.12	0.19	1.92
cpu1	FPexecMaint_125_1_1_FPexecMaint_0	6.431	1.411	1.76
cpu1	FileAgent_125_1_1_FileAgent_1	0.17	0.24	1.6
cpu1	FpUeIpHdlr_125_1_1_0_FpUeIpHdlr_0	6.431	1.411	1.76
cpu1	FpUeIpHdlr_125_1_1_1_FpUeIpHdlr_1	6.431	1.411	1.76
cpu1	GwStatsAgent_125_1_1_GwStatsAgent_1	0.12	0.26	1.92
cpu1	HwAgent_125_1_1_HwAgent_2	0.11	0.25	1.6
cpu1	ImAgent_125_1_1_ImAgent_1	0.1	0.18	1.6
cpu1	NRA_125_1_1_NRA_1	0.14	0.27	2.24
cpu1	NSAgent_125_1_1_NSAgent_0	0.12	0.22	1.92
cpu1	NmdAgent_125_1_1_NmdAgent_1	0.09	0.19	1.44
cpu1	SecPolicyAgt_125_1_1_SecPolicyAgt_0	6.431	1.411	1.92
cpu1	ShadowTask_ShadowTask_0	0.1	0.19	1.6
cpu1	TosfpFlowAgt_125_1_1_TosfpFlowAgt_0	6.431	1.411	0.48
cpu1	TosfpGwHdlr_125_1_1_0_TosfpGwHdlr_0	6.431	1.411	1.44
cpu1	TosfpGwHdlr_125_1_1_1_TosfpGwHdlr_1	6.431	1.411	1.6
cpu1	TosfpTblDist_TosfpTblDist_1	0.09	1.26	1.44
cpu1	TransProxyAgt_125_1_1_TransProxyAgt_0	6.431	1.411	1.6

```
cpu1      TransProxySipAlgAgt_125_1_1_TransProxySipAlgAgt_0      6.431      1.411      2.08
```

show cluster node <id> task operation-detail oper-state

Use the **show cluster node <id> task** command to display processes at the node level and show details to help identify issues related to a specific process.

Use the following parameters to display the operational status of tasks:

```
an3000-mcm-slot7cpu1# show cluster node {id} task operation-
detail oper-state [congested|down|dryup|maint|partially-
up|up]
```

For example:

```
an3000-mcm-slot7cpu1# show cluster node task operation-
detail oper-state
```

```
cluster
      CPU
OPER
ID   ID  NUMBER  TASK NAME
STATE
-----
----
125  1   cpu1     EdrTask_125_1_1_EdrTask_0                up
      cpu1     EventConfigRelayAgent_125_1_1_EventConfigRelayAgent_1 up
      cpu1     FMAgent_125_1_1_FMAgent_1                up
      cpu1     FPexecMaint_125_1_1_FPexecMaint_0          up
      cpu1     FileAgent_125_1_1_FileAgent_1              up
      cpu1     FpUeIpHdlr_125_1_1_0_FpUeIpHdlr_0          up
      cpu1     FpUeIpHdlr_125_1_1_1_FpUeIpHdlr_1          up
      cpu1     GwStatsAgent_125_1_1_GwStatsAgent_1         up
      cpu1     HwAgent_125_1_1_HwAgent_2                  up
      cpu1     ImAgent_125_1_1_ImAgent_1                  up
      cpu1     NRA_125_1_1_NRA_1                          up
      cpu1     NSAgent_125_1_1_NSAgent_0                  up
      cpu1     NmdAgent_125_1_1_NmdAgent_1                up
      cpu1     SecPolicyAgt_125_1_1_SecPolicyAgt_0         up
      cpu1     ShadowTask_ShadowTask_0                    up
      cpu1     TosfpFlowAgt_125_1_1_TosfpFlowAgt_0         up
      cpu1     TosfpGwHdlr_125_1_1_0_TosfpGwHdlr_0        up
      cpu1     TosfpGwHdlr_125_1_1_1_TosfpGwHdlr_1        up
      cpu1     TosfpTblDist_TosfpTblDist_1                up
      cpu1     TransProxyAgt_125_1_1_TransProxyAgt_0       up
      cpu1     TransProxySipAlgAgt_125_1_1_TransProxySipAlgAgt_0 up
```

show cluster statistics

Use the **show cluster statistics** command to display details of each VM's memory, CPU utilization and process count. Both CPU and memory should be no higher than 80% for the MCC to function predictably.

The SSM nodes dedicate a large percentage of their cores to packet handling loops to avoid certain performance penalties caused by using the Linux kernel. This command displays system status from a Linux perspective.

```
an3000-mcm-slot7cpu1# show cluster statistics
```

```
cluster
```

						ISOLATED		ISOLATED	
						CPU		CPU	
						UTILIZATION		UTILIZATION	
		CPU	CPU	MEMORY	PROCESS	UTILIZATION	GENERAL CPU	NORMALIZED	APPLICATION
ID	SLOT	NUMBER	UTILIZATION	UTILIZATION	COUNT	NORMALIZED	UTILIZATION	MAX	LOAD
125	1	1	64.55	33.39	213	00.00	04.98	00.00	02.60
	2	1	26.79	26.44	212	00.00	16.06	00.00	00.32
	3	1	63.91	32.43	212	00.00	01.69	00.00	01.13
	4	1	26.78	24.93	208	00.00	15.67	00.00	00.24
	5	1	0	0	0	0	0	0	0
	7	1	04.10	11.38	365	00.00	04.10	00.00	01.12
	8	1	01.37	35.58	215	00.00	01.37	00.00	00.22
	9	1	66.09	33.23	213	00.00	14.93	00.00	01.26
	12	1	26.42	35.09	212	00.00	15.73	00.00	00.24
	13	1	64.76	32.40	211	00.00	05.85	00.00	02.18
	14	1	26.36	24.20	212	00.00	15.38	00.00	00.25
	15	1	38.28	18.22	247	00.00	00.85	00.00	00.30

For more details, see *Slot Statistics* in *Acuitas Performance Statistics*.

Use the **show zone default system statistics** command to display CPU utilization that has been adjusted by the system to account for the internal architecture decisions, and displays a more realistic value of the CPU utilization.

show cluster control-plane protection status

Use the **control-plane-protection status** command to display the control plane redundancy table of communication of the VMs over the base network. The base network is responsible for inter-VM control communication; no data traffic is transferred on this path. Each VM has two redundant paths to the base network using port-A or port-B.

```
an3000-mcm-slot7cpu1# show cluster control-plane-protection
status
```

```
cluster

          BITMAP      BITMAP
          REACHABLE   REACHABLE
          NODES       NODES
ID  ID  USING  NUM  SWITCHOVERS  PORT A  PORT B  MOST RECENT SWITCHOVER CAUSE
-----
32  1    port-A  2          0xcf      0x0f      2020-01-21T14:44:20-0500 Switching to path:A
                               map: 00, (A=eth0, B=eth4), active MCM path A
                               Initialize to path:A map: [0-0],
                               (A=eth0, B=eth4), standby dev=eth4
      2    port-A  0          0xcf      0x0f
```

Note: The nodes on the system will be on path A (MCM-7) or path B (MCM-8) depending on which MCM is active.

show cluster data-plane protection status

Use the **data-plane-protection status** command to display the data plane redundancy table relating to VMs on the E-W (East-West traffic) network. Each VM has two redundant paths for E-W communication.

```
an3000-mcm-slot7cpu1# show cluster data-plane-protection
status
```

```
cluster

          BITMAP      BITMAP
          REACHABLE   REACHABLE
          NODES       NODES
ID  ID  USING  NUM  SWITCHOVERS  PORT A  PORT B  MOST RECENT SWITCHOVER CAUSE
-----
73  1    port-A  0          0x00      0x00      Initialize to path:A map: [0-0],
                               (A=kni0, B=), Redundancy disabled
      5    port-A  0          0x00      0x00      Initialize to path:A map: [0-0],
                               (A=fab0, B=), Redundancy disabled
```

show cluster control-plane-protection keepalive

Use the **keepalive** parameter with the **data-plane-protection** and **control-plane-protection** commands to display a matrix of the response times between each VM (NODE ID) and its cluster neighbor (PEER NODE ID) over both redundant paths.

Round Trip Times (RTT) greater than 300ms should be investigated. Extremely high RTT, for example, 15324 or a value of 0, usually indicates that communication on that link is broken and requires urgent attention.

```
an3000-mcm-slot7cpu1# show cluster control-plane-protection
keepalive
```

```
cluster

          5          5          5          5
          PORT A  MINUTE  MINUTE          PORT B  MINUTE  MINUTE
          PORT A  MOST    HISTORY  HISTORY  PORT B  MOST    HISTORY  HISTORY
          PEER  MOST    RECENT  PERCENT  PERCENT  MOST    RECENT  PERCENT  PERCENT
```

		NODE	NODE	RECENT	RX+TX	RX PORT	RX+TX	RECENT	RX TX	RX PORT	RX+TX
ID	ID	ID	ID	RX MS	MS	A	PORT A	RX MS	MS	B	PORT B

32	1	2		146	146	100	100	146	146	100	100
	1	3		1	1	100	100	202	202	100	100
	1	4		84	84	100	100	84	84	100	100
	1	7		125	125	100	100	0	0	0	0
	1	8		125	125	100	100	0	0	0	0
	2	1		30	30	100	100	30	30	100	100
	2	3		156	156	100	100	156	156	100	100
	2	4		38	38	100	100	38	38	100	100
	2	7		79	79	100	100	0	0	0	0
	2	8		79	79	100	100	0	0	0	0

show cluster data-plane-protection keepalive

```
an3000-mcm-slot7cpu1# show cluster data-plane-protection
keepalive
```

cluster											
				5		5		5		5	
				PORT A	MINUTE	MINUTE		PORT B		MINUTE	MINUTE
				PORT A	MOST	HISTORY	HISTORY	PORT B	MOST	HISTORY	HISTORY
				PEER	MOST	RECENT	PERCENT	PERCENT	MOST	RECENT	PERCENT
NODE		NODE	RECENT	RX+TX	RX	PORT	RX+TX	RECENT	RX+TX	RX	PORT
ID	ID	ID	RX MS	MS	A		PORT A	RX MS	MS	B	PORT B

125	1	2	116	116	100		100	116	116	100	100
	1	3	193	193	100		100	193	193	100	100
	1	4	5	5	100		100	5	5	100	100
	1	9	28	28	100		100	28	28	100	100
	1	12	172	172	100		100	172	172	100	100
	1	13	194	194	100		100	194	194	100	100
	1	14	142	142	100		100	142	142	100	100
	1	15	178	178	100		100	178	178	100	100

cpu1	FMAgent_125_1_1_FMAgent_1	0.19
cpu1	FPexecMaint_125_1_1_FPexecMaint_0	1.411
cpu1	FileAgent_125_1_1_FileAgent_1	0.24
cpu1	FpUeIpHdlr_125_1_1_0_FpUeIpHdlr_0	1.411
cpu1	FpUeIpHdlr_125_1_1_1_FpUeIpHdlr_1	1.411
cpu1	GwStatsAgent_125_1_1_GwStatsAgent_1	0.26
cpu1	HwAgent_125_1_1_HwAgent_2	0.25
cpu1	ImAgent_125_1_1_ImAgent_1	0.18
cpu1	NRA_125_1_1_NRA_1	0.27
cpu1	NSAgent_125_1_1_NSAgent_0	0.22
cpu1	NmdAgent_125_1_1_NmdAgent_1	0.19
cpu1	SecPolicyAgt_125_1_1_SecPolicyAgt_0	1.411
cpu1	ShadowTask_ShadowTask_0	0.19
cpu1	TosfpFlowAgt_125_1_1_TosfpFlowAgt_0	1.411
cpu1	TosfpGwHdlr_125_1_1_0_TosfpGwHdlr_0	1.411
cpu1	TosfpGwHdlr_125_1_1_1_TosfpGwHdlr_1	1.411
cpu1	TosfpTblDist_TosfpTblDist_1	1.26
cpu1	TransProxyAgt_125_1_1_TransProxyAgt_0	1.411
cpu1	TransProxySipAlgAgt_125_1_1_TransProxySipAlgAgt_0	1.411

an3000-mcm-slot7cpu1# **show cluster 125 node 1 task statistics cpu-utilization**

task statistics

CPU		CPU
NUMBER	TASK NAME	UTILIZATION

cpu1	EdrTask_125_1_1_EdrTask_0	6.43
cpu1	EventConfigRelayAgent_125_1_1_EventConfigRelayAgent_1	0.1
cpu1	FMAgent_125_1_1_FMAgent_1	0.12
cpu1	FPexecMaint_125_1_1_FPexecMaint_0	6.43
cpu1	FileAgent_125_1_1_FileAgent_1	0.17
cpu1	FpUeIpHdlr_125_1_1_0_FpUeIpHdlr_0	6.43
cpu1	FpUeIpHdlr_125_1_1_1_FpUeIpHdlr_1	6.43
cpu1	GwStatsAgent_125_1_1_GwStatsAgent_1	0.12
cpu1	HwAgent_125_1_1_HwAgent_2	0.1
cpu1	ImAgent_125_1_1_ImAgent_1	0.1
cpu1	NRA_125_1_1_NRA_1	0.14
cpu1	NSAgent_125_1_1_NSAgent_0	0.13
cpu1	NmdAgent_125_1_1_NmdAgent_1	0.09
cpu1	SecPolicyAgt_125_1_1_SecPolicyAgt_0	6.43
cpu1	ShadowTask_ShadowTask_0	0.1
cpu1	TosfpFlowAgt_125_1_1_TosfpFlowAgt_0	6.43
cpu1	TosfpGwHdlr_125_1_1_0_TosfpGwHdlr_0	6.43

```

cpu1    TosfpGwHdlr_125_1_1_1_TosfpGwHdlr_1          6.43
cpu1    TosfpTblDist_TosfpTblDist_1                  0.09
cpu1    TransProxyAgt_125_1_1_TransProxyAgt_0          6.43
cpu1    TransProxySipAlgAgt_125_1_1_TransProxySipAlgAgt_0 6.43

```

show cluster node fault-summary

When a software component of the MCC faults it generates a fault file, which is essentially a memory dump at the time of the fault. This file helps Affirmed Networks Support to identify and resolve the root cause of the issue. The fault file is stored in */public/corefiles/* which is accessible through the MCM Linux shell.

```

an3000-mcm-slot7cpu1# show cluster node fault-summary

cluster
    SECONDS
    SINCE
    ID ID EPOCH      FAULT FILENAME                                TIME
-----
73 7  1576606438  slot7cpu1/fault_8965_1576606438_sh.gz          Tue 17 Dec 2019 01:13:58 PM EST
    1576606444  slot7cpu1/fault_8719_1576606438_NmdServiceDir_7 Tue 17 Dec 2019 01:14:04 PM EST
    1576606455  slot7cpu1/fault_8966_1576606438_conf.d.gz       Tue 17 Dec 2019 01:14:15 PM EST
8  1576082869  slot8cpu1/fault_7403_1576082869_sh.gz          Wed 11 Dec 2019 11:47:49 AM EST
    1576082872  slot8cpu1/fault_6955_1576082869_NmdServiceDir_7.gz Wed 11 Dec 2019 11:47:52 AM EST
    1576082888  slot8cpu1/fault_7404_1576082869_conf.d.gz       Wed 11 Dec 2019 11:48:08 AM EST

```

show cluster node disk

Use the **disk** parameter to display the disk size, utilization, and error status of the VMs within the MCC cluster.

```

an3000-mcm-slot7cpu1# show cluster node disk

cluster
    ADMIN  OPER  DISK  DISK  USE  DISK
    ID  ID  DISKID  NAME      STATE  STATUS  SIZE  USAGE  PERCENT  ERROR
-----
73  1  1      proxy-cache  enabled up      -      -      -      No
    2      storage    enabled up      10.4GB 1.1GB  10%   No
    3      images     enabled up      8.3GB 18.9MB 0%    No
    4      rollback-image enabled up      10.4GB 6.1GB  58%   No
    5      prepared-image enabled up      10.4GB 23.7MB 0%    No
    6      operational-image enabled up      10.4GB 6.1GB  58%   No
    8      ram-storage disabled down    -      -      -      No

```

cluster <id> clear-faults

After collecting the fault files, use the **clear-faults** parameter to clear the files from the system.

For example:

```
an3000-mcm-slot7cpu1# cluster 11 clear-faults
```

Normally, you will also need to clear the associated alarm.

For example:

```
an3000-mcm-slot7cpu1# alarm active 223232 clear
```

cluster <id> reboot

Use the **cluster <id> reboot** command to reboot the entire virtual chassis. Issue this command with caution to avoid data loss.

For example:

```
an3000-mcm-slot7cpu1# cluster 11 reboot
```

cluster <id> node <node-id> reboot

Use the **cluster <id> node <node-id> reboot** command to reboot a specific node (VM) on the cluster.

For example:

```
an3000-mcm-slot7cpu1# cluster 11 node 3 reboot
```

cluster <id> node <node-id> processes

Use the parameters in [Cluster Node Processes Parameters](#) to perform the relevant action on the node processes.

For example:

```
an3000-mcm-slot7cpu1# cluster 11 node 2 processes cpu1
HttpProxy_100_42_1_138_HttpProxy_138 restart
```

Table 1-2 Cluster Node Processes Parameters

Parameter	Description
clearrestartdebug	Restart this application in normal mode after debugging
fault	Kill a task and dump its fault file
force-restart	Force restart this application
maintenance	Cause this process to stop accepting new work and shed existing work
ping	Ping a task
restart	Restart this application

Parameter	Description
restartdebug	Restart this application in debug mode

cluster <id> mcmswitchover

Maintenance activities or issues may require an MCM switchover, making the standby MCM active. Use the **cluster <id> mcmswitchover** command to perform a graceful handover between the two nodes.

For example:

```
an3000-mcm-slot7cpu1# cluster 12 mcmswitchover
```

System Commands

Use the system commands to display information related to the system.

show image-management release-modifier-status

Use the **show image-management release-modifier-status** command to display the DPI signature filename.

```
an3000-mcm-slot8cpu1# show image-management release-
modifier-status
```

```
image-management release-modifier-status
```

RELEASE ROLE	RELEASE VERSION	RELEASE MODIFIER	OPERATIONAL VERSION	ROLLBACK VERSION	ORIGINAL VERSION
operational	9.3.2.7-44.REL	dpisig	dpisig_2020-01-13-5	NotAvailable	dpisig_2019-12-13
operational	9.3.2.7-44.REL	maint	maint_2019-12-11	NotAvailable	maint_2017-06-12
operational	9.3.2.7-44.REL	plugin	plugin_2019-08-29	NotAvailable	plugin_2019-08-28
prepared	10.0.0.0-183.REL	dpisig	dpisig_2020-01-13-5	NotAvailable	dpisig_2020-01-13-5
prepared	10.0.0.0-183.REL	maint	maint_2019-12-11	NotAvailable	maint_2019-12-11
prepared	10.0.0.0-183.REL	plugin	plugin_2019-08-29	NotAvailable	plugin_2019-08-29

show infrastructure ntp-server status

Use the **show infrastructure ntp-server** command to display the health statistics of the connection between the MCC and the configured NTP servers.

```
an3000-mcm-slot7cpu1# show infrastructure ntp-server status
```

```
infrastructure ntp-server status
```

REMOTE	SYNC	REFID	ST	T	WHEN	POLL	REACH	DELAY	OFFSET	JITTER
10.32.0.254	*	216.229.4.69	3	u	103	128	377	0.212	3.226	0.040
master		10.32.0.254	4	u	72	16	0	0.141	-0.560	0.000

show infrastructure file-management statistics

Use the **show infrastructure file-management statistics** command to display the file management activities on the MCC. Pay careful attention to the FILES PURGED DISK FULL column because an incremented value indicates a full file system and possible irretrievable data loss.

```
an3000-mcm-slot7cpu1# show infrastructure file-management
statistics
```

```
infrastructure file-management statistics
```

PREFIX	FILES	TRANSFERRED	FILES	PURGED	DISK	FULL

```

-----
//boot/partitions/storage/*          0          0      0
//boot/partitions/storage/dump/*      0          0      0
//var/log/asp/*                       0          126454  0
//var/log/psm/*                       0          0      0
/var/corefiles/*                      0          0      0
/var/log/datarecord.lnk/HTDR_         0          0      0
/var/log/eventlog.lnk/*               0          0      0
/var/log/eventlog.lnk/IMSITRACE_PCAPS 0          0      0
/var/log/eventlog.lnk/critical        0          0      0

```

show zone default system statistics

Use the **show zone default system statistics** command to display an extensive system level overview of the statistics of the VMs within the cluster. Use the **core-level**, **cpu-level**, and **node-level** parameters to display a more manageable subset of the required output.

For example:

```
an3000-mcm-slot7cpu1# show zone default system statistics
```

```

system statistics node-level cpu-stats

```

		AVG	AVG	AVG	AVG	AVG	AVG	AVG	AVG	AVG
		1MIN	5MIN	15MIN	1MIN	5MIN	15MIN	1MIN	5MIN	15MIN
		CSM	CSM	CSM	SSM	SSM	SSM	MCM	MCM	MCM
		CPU	CPU	CPU	CPU	CPU	CPU	CPU	CPU	CPU
CHASSIS	SLOT	UTIL	UTIL	UTIL	UTIL	UTIL	UTIL	UTIL	UTIL	UTIL
73	1	17	17	17	0	0	0	0	0	0
73	5	0	0	0	0	0	0	0	0	0
73	7	0	0	0	0	0	0	4	3	3
73	8	0	0	0	0	0	0	2	2	2

```

-----
system statistics node-level memory-stats 73 1
Average 1 minute csm memory utilization          41
Average 5 minute csm memory utilization          41
Average 15 minute csm memory utilization         41
Average 1 minute ssm memory utilization          0
Average 5 minute ssm memory utilization          0
Average 15 minute ssm memory utilization         0
Average 1 minute mcm memory utilization          0
Average 5 minute mcm memory utilization          0
Average 15 minute mcm memory utilization         0
Total available memory                          32682636
Total available memory free                     18970520
Total available memory in buffer cache           140944
Total available memory in page cache minus swap cache 3701988
Total available memory in swap cache             0

```

Diagnosing Issues and Performing Health Checks

Total available virtual memory active	10909688
Total available virtual memory inactive	3612332
Total available swap memory	0
Total available swap memory free	0
Total available memory that needs to be written to disk or swap	28
Total available memory which is actively being written back to the disk	0
Total Non-file backed pages mapped into userspace page tables	10685924
Total available memory that has been mapped (mmap)	3526400
Total available memory in the slab (kernel data structures)	115284
Total available memory part of slab, that might be reclaimed, such as caches	50188
Total available memory part of slab that cannot be reclaimed on memory pressure	65096
Total available memory dedicated to lowest level page tables	160292
Total available memory NFS pages sent to the server, but not yet committed to stable storage	0
Total available memory block device bounce buffers	0
Total available memory used by FUSE for temporary writeback buffers	0
Total available memory commit limit	15112516
Total available memory presently allocated	23027832
Total available virtual memory	34359738367
Total available virtual memory used	195128
Largest available virtual memory chunk	34359423904
Total available huge tlb pages	1200
Total available huge tlb pages free	0
Total available huge tlb pages reserved	0
Total available huge tlb pages surplus	0
Total available huge tlb page size	2048
system statistics node-level memory-stats 73 5	
Average 1 minute csm memory utilization	0
Average 5 minute csm memory utilization	0
Average 15 minute csm memory utilization	0
Average 1 minute ssm memory utilization	33
Average 5 minute ssm memory utilization	33
Average 15 minute ssm memory utilization	33
Average 1 minute mcm memory utilization	0
Average 5 minute mcm memory utilization	0
Average 15 minute mcm memory utilization	0
Total available memory	32682636
Total available memory free	21698760
Total available memory in buffer cache	142908
Total available memory in page cache minus swap cache	386520
Total available memory in swap cache	0
Total available virtual memory active	5768812
Total available virtual memory inactive	299048
Total available swap memory	0
Total available swap memory free	0
Total available memory that needs to be written to disk or swap	12
Total available memory which is actively being written back to the disk	0
Total Non-file backed pages mapped into userspace page tables	5548492

Total available memory that has been mapped (mmap)	215336
Total available memory in the slab (kernel data structures)	56416
Total available memory part of slab, that might be reclaimed, such as caches	23492
Total available memory part of slab that cannot be reclaimed on memory pressure	32924
Total available memory dedicated to lowest level page tables	25400
Total available memory NFS pages sent to the server, but not yet committed to stable storage	0
Total available memory block device bounce buffers	0
Total available memory used by FUSE for temporary writeback buffers	0
Total available memory commit limit	13781316
Total available memory presently allocated	7664940
Total available virtual memory	34359738367
Total available virtual memory used	191720
Largest available virtual memory chunk	34359422896
Total available huge tlb pages	2500
Total available huge tlb pages free	0
Total available huge tlb pages reserved	0
Total available huge tlb pages surplus	0
Total available huge tlb page size	2048
system statistics node-level memory-stats 73 7	
Average 1 minute csm memory utilization	0
Average 5 minute csm memory utilization	0
Average 15 minute csm memory utilization	0
Average 1 minute ssm memory utilization	0
Average 5 minute ssm memory utilization	0
Average 15 minute ssm memory utilization	0
Average 1 minute mcm memory utilization	36
Average 5 minute mcm memory utilization	36
Average 15 minute mcm memory utilization	36
Total available memory	7910136
Total available memory free	5030504
Total available memory in buffer cache	239744
Total available memory in page cache minus swap cache	4249864
Total available memory in swap cache	0
Total available virtual memory active	5299548
Total available virtual memory inactive	1590980
Total available swap memory	0
Total available swap memory free	0
Total available memory that needs to be written to disk or swap	524
Total available memory which is actively being written back to the disk	0
Total Non-file backed pages mapped into userspace page tables	2400968
Total available memory that has been mapped (mmap)	125772
Total available memory in the slab (kernel data structures)	237300
Total available memory part of slab, that might be reclaimed, such as caches	162220
Total available memory part of slab that cannot be reclaimed on memory pressure	75080
Total available memory dedicated to lowest level page tables	38488
Total available memory NFS pages sent to the server, but not yet committed to stable storage	0
Total available memory block device bounce buffers	0

Total available memory used by FUSE for temporary writeback buffers	0
Total available memory commit limit	3955068
Total available memory presently allocated	4646072
Total available virtual memory	34359738367
Total available virtual memory used	151760
Largest available virtual memory chunk	34359455728
Total available huge tlb pages	0
Total available huge tlb pages free	0
Total available huge tlb pages reserved	0
Total available huge tlb pages surplus	0
Total available huge tlb page size	2048
system statistics node-level memory-stats 73 8	
Average 1 minute csm memory utilization	0
Average 5 minute csm memory utilization	0
Average 15 minute csm memory utilization	0
Average 1 minute ssm memory utilization	0
Average 5 minute ssm memory utilization	0
Average 15 minute ssm memory utilization	0
Average 1 minute mcm memory utilization	28
Average 5 minute mcm memory utilization	28
Average 15 minute mcm memory utilization	28
Total available memory	7910136
Total available memory free	5673224
Total available memory in buffer cache	205236
Total available memory in page cache minus swap cache	4949120
Total available memory in swap cache	0
Total available virtual memory active	5774600
Total available virtual memory inactive	1187512
Total available swap memory	0
Total available swap memory free	0
Total available memory that needs to be written to disk or swap	8
Total available memory which is actively being written back to the disk	0
Total Non-file backed pages mapped into userspace page tables	1807624
Total available memory that has been mapped (mmap)	165284
Total available memory in the slab (kernel data structures)	252592
Total available memory part of slab, that might be reclaimed, such as caches	210628
Total available memory part of slab that cannot be reclaimed on memory pressure	41964
Total available memory dedicated to lowest level page tables	32388
Total available memory NFS pages sent to the server, but not yet committed to stable storage	0
Total available memory block device bounce buffers	0
Total available memory used by FUSE for temporary writeback buffers	0
Total available memory commit limit	3955068
Total available memory presently allocated	3467084
Total available virtual memory	34359738367
Total available virtual memory used	151568
Largest available virtual memory chunk	34359468068
Total available huge tlb pages	0

```

Total available huge tlb pages free                                0
Total available huge tlb pages reserved                          0
Total available huge tlb pages surplus                          0
Total available huge tlb page size                              2048

system statistics node-level pfm-stats 73 1
Average 1 minute Disk write speed MB per second    142
Average 5 minute Disk write speed MB per second    159
Average 15 minute Disk write speed MB per second   169
Average 1 minute Base Rx Mbps per second           0
Average 1 minute Base Tx Mbps per second           0
Average 5 minute Base RxMbps per second            0
Average 5 minute Base Tx Mbps per second           0
Average 15 minute Base Rx Mbps per second          0
Average 15 minute Base Tx Mbps per second          0
Average 1 minute wc ping delay usec                184
Average 5 minute wc ping delay usec                1388
Average 15 minute wc ping delay usec               669
Ping worse case delay node                          7
Average 1 minute queue length bytes                0
Ping worse case delay port                        8142
Average 15 minute Base link monitor drops path A 0
Average 15 minute Base link monitor drops path B 0
Cumulative base delay 15 minute interval ms        0
Average 15 minute Data link monitor drops path A 0
Average 15 minute Data link monitor drops path B 0
Cumulative backlog queue drops 15 minute           0

system statistics node-level pfm-stats 73 5
Average 1 minute Disk write speed MB per second    171
Average 5 minute Disk write speed MB per second    156
Average 15 minute Disk write speed MB per second   165
Average 1 minute Base Rx Mbps per second           0
Average 1 minute Base Tx Mbps per second           0
Average 5 minute Base RxMbps per second            0
Average 5 minute Base Tx Mbps per second           0
Average 15 minute Base Rx Mbps per second          0
Average 15 minute Base Tx Mbps per second          0
Average 1 minute wc ping delay usec                149
Average 5 minute wc ping delay usec                137
Average 15 minute wc ping delay usec               136
Ping worse case delay node                          7
Average 1 minute queue length bytes                0
Ping worse case delay port                        8151
Average 15 minute Base link monitor drops path A 0
Average 15 minute Base link monitor drops path B 0
Cumulative base delay 15 minute interval ms        0
Average 15 minute Data link monitor drops path A 0
Average 15 minute Data link monitor drops path B 0

```

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```
Cumulative backlog queue drops 15 minute      0
system statistics node-level pfm-stats 73 7
Average 1 minute Disk write speed MB per second 141
Average 5 minute Disk write speed MB per second 138
Average 15 minute Disk write speed MB per second 139
Average 1 minute Base Rx Mbps per second      0
Average 1 minute Base Tx Mbps per second      0
Average 5 minute Base RxMbps per second       0
Average 5 minute Base Tx Mbps per second      0
Average 15 minute Base Rx Mbps per second     0
Average 15 minute Base Tx Mbps per second     0
Average 1 minute wc ping delay usec           180
Average 5 minute wc ping delay usec           158
Average 15 minute wc ping delay usec          158
Ping worse case delay node                    1
Average 1 minute queue length bytes           0
Ping worse case delay port                     8147
Average 15 minute Base link monitor drops path A 0
Average 15 minute Base link monitor drops path B 0
Cumulative base delay 15 minute interval ms   0
Average 15 minute Data link monitor drops path A 0
Average 15 minute Data link monitor drops path B 0
Cumulative backlog queue drops 15 minute      0
system statistics node-level pfm-stats 73 8
Average 1 minute Disk write speed MB per second 157
Average 5 minute Disk write speed MB per second 156
Average 15 minute Disk write speed MB per second 157
Average 1 minute Base Rx Mbps per second      0
Average 1 minute Base Tx Mbps per second      0
Average 5 minute Base RxMbps per second       0
Average 5 minute Base Tx Mbps per second      0
Average 15 minute Base Rx Mbps per second     0
Average 15 minute Base Tx Mbps per second     0
Average 1 minute wc ping delay usec           156
Average 5 minute wc ping delay usec           154
Average 15 minute wc ping delay usec          155
Ping worse case delay node                    7
Average 1 minute queue length bytes           0
Ping worse case delay port                     8105
Average 15 minute Base link monitor drops path A 0
Average 15 minute Base link monitor drops path B 0
Cumulative base delay 15 minute interval ms   0
Average 15 minute Data link monitor drops path A 0
Average 15 minute Data link monitor drops path B 0
Cumulative backlog queue drops 15 minute      0
```

system statistics cpu-level cpu-stats

		AVG	AVG	AVG	AVG	AVG	AVG	AVG	AVG	AVG	AVG
		1MIN	5MIN	15MIN	1MIN	5MIN	15MIN	1MIN	5MIN	15MIN	
		CSM	CSM	CSM	SSM	SSM	SSM	MCM	MCM	MCM	
		CPU	CPU	CPU	CPU	CPU	CPU	CPU	CPU	CPU	
CHASSIS	SLOT	CPU	UTIL	UTIL	UTIL	UTIL	UTIL	UTIL	UTIL	UTIL	UTIL

73	1	1	17	17	17	0	0	0	0	0	0
73	5	1	0	0	0	0	0	0	0	0	0
73	7	1	0	0	0	0	0	0	4	3	3
73	8	1	0	0	0	0	0	0	2	2	2

system statistics cpu-level memory-stats 73 1 1

Average 1 minute csm memory utilization	41
Average 5 minute csm memory utilization	41
Average 15 minute csm memory utilization	41
Average 1 minute ssm memory utilization	0
Average 5 minute ssm memory utilization	0
Average 15 minute ssm memory utilization	0
Average 1 minute mcm memory utilization	0
Average 5 minute mcm memory utilization	0
Average 15 minute mcm memory utilization	0
Total available memory	32682636
Total available memory free	18970520
Total available memory in buffer cache	140944
Total available memory in page cache minus swap cache	3701988
Total available memory in swap cache	0
Total available virtual memory active	10909688
Total available virtual memory inactive	3612332
Total available swap memory	0
Total available swap memory free	0
Total available memory that needs to be written to disk or swap	28
Total available memory which is actively being written back to the disk	0
Total Non-file backed pages mapped into userspace page tables	10685924
Total available memory that has been mapped (mmap)	3526400
Total available memory in the slab (kernel data structures)	115284
Total available memory part of slab, that might be reclaimed, such as caches	50188
Total available memory part of slab that cannot be reclaimed on memory pressure	65096
Total available memory dedicated to lowest level page tables	160292
Total available memory NFS pages sent to the server, but not yet committed to stable storage	0
Total available memory block device bounce buffers	0
Total available memory used by FUSE for temporary writeback buffers	0
Total available memory commit limit	15112516
Total available memory presently allocated	23027832
Total available virtual memory	34359738367
Total available virtual memory used	195128
Largest available virtual memory chunk	34359423904
Total available huge tlb pages	1200

Total available huge tlb pages free	0
Total available huge tlb pages reserved	0
Total available huge tlb pages surplus	0
Total available huge tlb page size	2048
system statistics cpu-level memory-stats 73 5 1	
Average 1 minute csm memory utilization	0
Average 5 minute csm memory utilization	0
Average 15 minute csm memory utilization	0
Average 1 minute ssm memory utilization	33
Average 5 minute ssm memory utilization	33
Average 15 minute ssm memory utilization	33
Average 1 minute mcm memory utilization	0
Average 5 minute mcm memory utilization	0
Average 15 minute mcm memory utilization	0
Total available memory	32682636
Total available memory free	21698760
Total available memory in buffer cache	142908
Total available memory in page cache minus swap cache	386520
Total available memory in swap cache	0
Total available virtual memory active	5768812
Total available virtual memory inactive	299048
Total available swap memory	0
Total available swap memory free	0
Total available memory that needs to be written to disk or swap	12
Total available memory which is actively being written back to the disk	0
Total Non-file backed pages mapped into userspace page tables	5548492
Total available memory that has been mapped (mmap)	215336
Total available memory in the slab (kernel data structures)	56416
Total available memory part of slab, that might be reclaimed, such as caches	23492
Total available memory part of slab that cannot be reclaimed on memory pressure	32924
Total available memory dedicated to lowest level page tables	25400
Total available memory NFS pages sent to the server, but not yet committed to stable storage	0
Total available memory block device bounce buffers	0
Total available memory used by FUSE for temporary writeback buffers	0
Total available memory commit limit	13781316
Total available memory presently allocated	7664940
Total available virtual memory	34359738367
Total available virtual memory used	191720
Largest available virtual memory chunk	34359422896
Total available huge tlb pages	2500
Total available huge tlb pages free	0
Total available huge tlb pages reserved	0
Total available huge tlb pages surplus	0
Total available huge tlb page size	2048
system statistics cpu-level memory-stats 73 7 1	
Average 1 minute csm memory utilization	0
Average 5 minute csm memory utilization	0

Average 15 minute csm memory utilization	0
Average 1 minute ssm memory utilization	0
Average 5 minute ssm memory utilization	0
Average 15 minute ssm memory utilization	0
Average 1 minute mcm memory utilization	36
Average 5 minute mcm memory utilization	36
Average 15 minute mcm memory utilization	36
Total available memory	7910136
Total available memory free	5030504
Total available memory in buffer cache	239744
Total available memory in page cache minus swap cache	4249864
Total available memory in swap cache	0
Total available virtual memory active	5299548
Total available virtual memory inactive	1590980
Total available swap memory	0
Total available swap memory free	0
Total available memory that needs to be written to disk or swap	524
Total available memory which is actively being written back to the disk	0
Total Non-file backed pages mapped into userspace page tables	2400968
Total available memory that has been mapped (mmap)	125772
Total available memory in the slab (kernel data structures)	237300
Total available memory part of slab, that might be reclaimed, such as caches	162220
Total available memory part of slab that cannot be reclaimed on memory pressure	75080
Total available memory dedicated to lowest level page tables	38488
Total available memory NFS pages sent to the server, but not yet committed to stable storage	0
Total available memory block device bounce buffers	0
Total available memory used by FUSE for temporary writeback buffers	0
Total available memory commit limit	3955068
Total available memory presently allocated	4646072
Total available virtual memory	34359738367
Total available virtual memory used	151760
Largest available virtual memory chunk	34359455728
Total available huge tlb pages	0
Total available huge tlb pages free	0
Total available huge tlb pages reserved	0
Total available huge tlb pages surplus	0
Total available huge tlb page size	2048
system statistics cpu-level memory-stats 73 8 1	
Average 1 minute csm memory utilization	0
Average 5 minute csm memory utilization	0
Average 15 minute csm memory utilization	0
Average 1 minute ssm memory utilization	0
Average 5 minute ssm memory utilization	0
Average 15 minute ssm memory utilization	0
Average 1 minute mcm memory utilization	28
Average 5 minute mcm memory utilization	28
Average 15 minute mcm memory utilization	28

Diagnosing Issues and Performing Health Checks

```

Total available memory                                7910136
Total available memory free                          5673224
Total available memory in buffer cache                205236
Total available memory in page cache minus swap cache 4949120
Total available memory in swap cache                  0
Total available virtual memory active                  5774600
Total available virtual memory inactive                1187512
Total available swap memory                           0
Total available swap memory free                      0
Total available memory that needs to be written to disk or swap 8
Total available memory which is actively being written back to the disk 0
Total Non-file backed pages mapped into userspace page tables 1807624
Total available memory that has been mapped (mmap)    165284
Total available memory in the slab (kernel data structures) 252592
Total available memory part of slab, that might be reclaimed, such as caches 210628
Total available memory part of slab that cannot be reclaimed on memory pressure 41964
Total available memory dedicated to lowest level page tables 32388
Total available memory NFS pages sent to the server, but not yet committed to stable storage 0
Total available memory block device bounce buffers    0
Total available memory used by FUSE for temporary writeback buffers 0
Total available memory commit limit                  3955068
Total available memory presently allocated             3467084
Total available virtual memory                       34359738367
Total available virtual memory used                   151568
Largest available virtual memory chunk                34359468068
Total available huge tlb pages                        0
Total available huge tlb pages free                   0
Total available huge tlb pages reserved               0
Total available huge tlb pages surplus                0
Total available huge tlb page size                    2048

```

system statistics core-level cpu-stats

				AVG	AVG	AVG	AVG	AVG	AVG	AVG	AVG	AVG
				1MIN	5MIN	15MIN	1MIN	5MIN	15MIN	1MIN	5MIN	15MIN
				CSM	CSM	CSM	SSM	SSM	SSM	MCM	MCM	MCM
				CPU	CPU	CPU	CPU	CPU	CPU	CPU	CPU	CPU
CHASSIS	SLOT	CPU	CORE	UTIL	UTIL	UTIL	UTIL	UTIL	UTIL	UTIL	UTIL	UTIL
73	1	1	1	7	6	6	0	0	0	0	0	0
73	1	1	2	6	5	5	0	0	0	0	0	0
73	1	1	3	0	0	0	0	0	0	0	0	0
73	1	1	4	100	100	100	0	0	0	0	0	0
73	1	1	5	7	6	7	0	0	0	0	0	0
73	1	1	6	7	6	6	0	0	0	0	0	0
73	1	1	7	7	6	6	0	0	0	0	0	0
73	1	1	8	6	7	6	0	0	0	0	0	0
73	5	1	1	0	0	0	1	1	1	0	0	0
73	5	1	2	0	0	0	2	1	1	0	0	0

```

73      5      1      3      0      0      0      0      0      0      0      0      0
73      5      1      4      0      0      0      0      0      0      0      0      0
73      5      1      5      0      0      0      0      0      0      0      0      0
73      5      1      6      0      0      0      0      0      0      0      0      0
73      5      1      7      0      0      0      0      0      0      0      0      0

```

```

system statistics intf-level intf-stats
                        AVG      AVG      AVG
                        1MIN     5MIN     15MIN
                        INTF     INTF     INTF
CHASSIS  SLOT  CPU   INTF  UTIL  UTIL  UTIL
-----
73        1      1      0      0      0      0
73        1      1      1      0      0      0
73        1      1      2      0      0      0
73        1      1      3      0      0      0
73        1      1      4      0      0      0
73        1      1      5      0      0      0
73        1      1      6      0      0      0
73        1      1      7      0      0      0
73        1      1      8      0      0      0
73        1      1      9      0      0      0
73        1      1     10      0      0      0

```

Using Log Files

You can retrieve log files from the MCC via the Linux shell for analysis. However, you can also view the non-packet capture files in `/var/log/eventlog.lnk` from the CLI. Navigate to the folder and press Tab to display a list of available filenames.

Note: Customized logs/debugs are stored by default in the eventlog directory.

```
an3000-mcm-slot7cpu1# show logs logfile <name>
```

TOS logs, which are binary by default, can also be viewed when archived from the Linux shell by unzipping the file and using the logconvert utility to convert the binary file to ASCII.

```

root@an3000-mcm-slot8cpu1:~# cp tos20150419133044_33.bin.gz /tmp/
root@an3000-mcm-slot8cpu1:~# cd /tmp/
root@an3000-mcm-slot8cpu1:~# gunzip tos20150419133044_33.bin.gz
root@an3000-mcm-slot8cpu1:~# logconvert tos20150419133044_33.bin
root@an3000-mcm-slot8cpu1:~# cat tos20150419133044_33.bin.txt

```

GGSN/PGW Commands

Use the commands and parameters in this section to view gateway information and related statistics.

Call Performance Tables

```
an3000-mcm-slot7cpu1# show zone default gateway statistics
cluster-level call-performance pgw <table-name>
```

current-sessions	1
bearers	1
create-session-attempts	80
create-session-accepts	80
create-session-rejects	0
create-session-rejects-malformed-requests	0
create-session-rejects-unknown-apn	0
create-session-rejects-resource-unavailable	0
create-session-rejects-service-not-supported	0
create-session-rejects-system-failure	0
create-session-requests-throttled-cluster-rate-control	0
create-session-requests-throttled-apn-rate-control	0
create-session-rejects-apn-in-congestion-state	0
create-session-rejects-apn-in-critical-state	0
create-session-rejects-apn-in-major-state	0
create-session-rejects-apn-in-minor-state	0
create-session-rejects-misc	0
create-session-rejects-change-session-to-colocated-failed	0
create-session-lapi-pdn-attempts	0
create-session-lapi-pdn-accepts	0
create-session-lapi-pdn-rejects	0
delete-session-attempts	1
delete-session-accepts	1
delete-session-rejects	0
ue-initiated-dedicated-bearer-activation-attempts	0
ue-initiated-dedicated-bearer-activation-accepts	0
ue-initiated-dedicated-bearer-activation-rejects	0
pgw-initiated-dedicated-bearer-activation-attempts	0
pgw-initiated-dedicated-bearer-activation-accepts	0
pgw-initiated-dedicated-bearer-activation-partial-accepts	0
pgw-initiated-dedicated-bearer-activation-rejects	0
mme-initiated-dedicated-bearer-deactivation-attempts	0
mme-initiated-dedicated-bearer-deactivation-accepts	0
mme-initiated-dedicated-bearer-deactivation-rejects	0
ue-initiated-dedicated-bearer-deactivation-attempts	0
ue-initiated-dedicated-bearer-deactivation-accepts	0
ue-initiated-dedicated-bearer-deactivation-rejects	0
pgw-initiated-dedicated-bearer-deactivation-attempts	0

pgw-initiated-dedicated-bearer-deactivation-accepts	0
pgw-initiated-dedicated-bearer-deactivation-rejects	0
modify-bearer-command-attempts	0
modify-bearer-command-rejects	0
update-bearer-attempts	0
update-bearer-accepts	0
update-bearer-partial-accepts	0
update-bearer-rejects	0
modify-bearer-attempts	79
modify-bearer-accepts	79
modify-bearer-rejects	0
modify-bearer-rejects-context-not-found	0
create-pdp-context-attempts	0
create-pdp-context-accepts	0
create-pdp-context-rejects	0
create-pdp-context-rejects-malformed-requests	0
create-pdp-context-rejects-unknown-apn	0
create-pdp-context-rejects-resource-unavailable	0
create-pdp-context-rejects-service-not-supported	0
create-pdp-context-rejects-system-failure	0
create-pdp-context-requests-throttled-cluster-rate-control	0
create-pdp-context-requests-throttled-apn-rate-control	0
create-pdp-context-rejects-misc	0
delete-pdp-context-attempts	0
delete-pdp-context-accepts	0
delete-pdp-context-rejects	0
ue-initiated-secondary-pdp-activation-attempts	0
ue-initiated-secondary-pdp-activation-accepts	0
ue-initiated-secondary-pdp-activation-rejects	0
ggsn-initiated-secondary-pdp-activation-attempts	0
ggsn-initiated-secondary-pdp-activation-accepts	0
ggsn-initiated-secondary-pdp-activation-rejects	0
sgsn-initiated-secondary-pdp-deactivation-attempts	0
sgsn-initiated-secondary-pdp-deactivation-accepts	0
sgsn-initiated-secondary-pdp-deactivation-rejects	0
ue-initiated-secondary-pdp-deactivation-attempts	0
ue-initiated-secondary-pdp-deactivation-accepts	0
ue-initiated-secondary-pdp-deactivation-rejects	0
ggsn-initiated-secondary-pdp-deactivation-attempts	0
ggsn-initiated-secondary-pdp-deactivation-accepts	0
ggsn-initiated-secondary-pdp-deactivation-rejects	0
sgsn-initiated-update-pdp-attempts	0
sgsn-initiated-update-pdp-accepts	0
sgsn-initiated-update-pdp-rejects	0
sgsn-initiated-update-pdp-rejects-context-not-found	0
ggsn-initiated-update-pdp-attempts	0
ggsn-initiated-update-pdp-accepts	0

ggsn-initiated-update-pdp-rejects	0
local-close-attempts	78
local-close-attempts-accepted	78
pcrf-initiated-close-attempts	0
ocs-initiated-close-attempts	0
idle-session-duration-exceeded-closes	0
maximal-session-duration-exceeded-closes	0
operator-initiated-pdn-session-closes	77
bearer-idle-duration-exceeded-deactivations	0
active-sessions	0
idle-sessions	1
current-geosync-restored-sessions	0
interval-peak-sessions	1
interval-peak-active-sessions	1
interval-peak-idle-sessions	1
default-bearers	1
dedicated-bearers	0
interval-peak-bearers	1
interval-peak-default-bearers	1
interval-peak-dedicated-bearers	0
local-close-attempts-dhcp-release	0
trace-session-activations	0
trace-session-activation-accepts	0
trace-session-deactivations	0
trace-session-deactivation-accepts	0
local-close-attempts-dhcp-timeout	0
local-close-attempts-dhcp-lease-expiry	0
local-close-radius-disconnect-closes	0
current-sessions-rat-type-utran	0
current-sessions-rat-type-geran	0
current-sessions-rat-type-wlan	0
current-sessions-rat-type-gan	0
current-sessions-rat-type-hspa-evolution	0
current-sessions-rat-type-e-utran	1
current-sessions-rat-type-virtual	0
current-sessions-gnnp-geran	0
current-sessions-gnnp-utran	0
current-sessions-s5s8-geran	0
current-sessions-s5s8-utran	0
current-sessions-roaming	0
current-sessions-rat-type-nbiot-eutran	0
current-sessions-using-gz	0
current-unique-subscribers	1
current-ipv4-pdn-sessions	1
current-ipv6-pdn-sessions	0
current-dual-stack-pdn-sessions	0
current-nonip-pdn-sessions	0

number-of-rat-session-handovers	0
number-of-current-gx-sessions	0
number-of-sessions-setup-with-default-gx-policy	60
number-of-active-sessions-with-default-gx-policy	1
number-of-active-gx-sessions-awaiting-pcrf-rejoin	1
number-of-successful-gx-pcrf-rejoin	0
number-of-current-gy-sessions	0
number-of-sessions-setup-with-default-gy-quota	0
number-of-active-sessions-setup-with-default-gy-quota	0
number-of-active-sessions-using-default-quota-triggered-by-ccru-failure	0
number-of-total-sessions-using-default-quota-triggered-by-ccru-failure	0
number-of-active-gy-sessions-awaiting-ocs-rejoin	0
number-of-successful-ocs-gy-rejoin	0
number-of-times-default-gy-quota-reapplied-ccri-failure	0
number-of-times-default-gy-quota-applied-ccru-failure	0
number-of-times-gy-down-gy-rejoin-timer-restarted	0
number-of-gy-records-write-to-disk-by-ccrt-failure	0
number-of-total-sessions-using-default-quota-triggered-by-3xx	0
number-of-total-sessions-using-default-quota-triggered-by-4xx	0
number-of-total-sessions-using-default-quota-triggered-by-5xx	0
number-of-total-sessions-using-default-quota-triggered-by-time-out	0
number-of-total-sessions-using-default-quota-triggered-by-ocs-link-down	0
number-of-total-sessions-using-default-quota-triggered-by-internal-error	0
local-close-attempts-duplicate-session	1
number-of-control-teid-lookup-failures	55
number-of-error-indications-processed	0
local-close-attempts-received-error-indication	0
dedicated-bearer-deletion-due-to-error-indication	0
number-of-s5u-default-bearer-error-indications-processed	0
number-of-s5u-default-bearer-local-close-attempts-received-error-indication	0
number-of-s5u-dedicated-bearer-error-indications-processed	0
number-of-s5u-dedicated-bearer-deactivation-due-to-error-indication	0
local-close-attempts-teid-collision	0
change-notification-attempts	0
change-notification-accepts	0
change-notification-rejects	0
change-notification-rejects-malformed-requests	0
change-notification-rejects-unknown-apn	0
change-notification-rejects-resource-unavailable	0
change-notification-rejects-service-not-supported	0
change-notification-rejects-system-failure	0
change-notification-rejects-misc	0
change-notification-req-drop	0
num-gtpp-cdr-sent	0
num-gtpp-cdr-resp-success	0
num-gtpp-cdr-resp-failure	0
num-gtpp-cdr-resp-timeout	0

bearers-setup-success	80
bearers-setup-fail	0
current-bearers-qci1	0
current-bearers-qci2	0
current-bearers-qci3	0
current-bearers-qci4	0
current-bearers-qci5	0
current-bearers-qci6	0
current-bearers-qci7	0
current-bearers-qci8	0
current-bearers-qci9	1
current-bearers-qci-other	0
bearers-setup-success-qci1	20
bearers-setup-success-qci2	0
bearers-setup-success-qci3	0
bearers-setup-success-qci4	0
bearers-setup-success-qci5	0
bearers-setup-success-qci6	0
bearers-setup-success-qci7	0
bearers-setup-success-qci8	0
bearers-setup-success-qci9	60
bearers-setup-success-qci-other	0
bearers-setup-fail-qci1	0
bearers-setup-fail-qci2	0
bearers-setup-fail-qci3	0
bearers-setup-fail-qci4	0
bearers-setup-fail-qci5	0
bearers-setup-fail-qci6	0
bearers-setup-fail-qci7	0
bearers-setup-fail-qci8	0
bearers-setup-fail-qci9	0
bearers-setup-fail-qci-other	0
saegw-pgw-current-sessions	1
saegw-pgw-current-sessions-s5s8-geran	0
saegw-pgw-current-sessions-s5s8-utran	0
saegw-pgw-current-sessions-eutran	1
saegw-pgw-current-sessions-nbiot-eutran	0
saegw-pgw-current-bearers	1
saegw-pgw-bearers-setup-success	80
saegw-pgw-bearers-setup-fail	0
saegw-pgw-current-bearers-qci1	0
saegw-pgw-current-bearers-qci2	0
saegw-pgw-current-bearers-qci3	0
saegw-pgw-current-bearers-qci4	0
saegw-pgw-current-bearers-qci5	0
saegw-pgw-current-bearers-qci6	0
saegw-pgw-current-bearers-qci7	0

saegw-pgw-current-bearers-qci8	0
saegw-pgw-current-bearers-qci9	1
saegw-pgw-current-bearers-qci-other	0
saegw-pgw-bearers-setup-success-qci1	20
saegw-pgw-bearers-setup-success-qci2	0
saegw-pgw-bearers-setup-success-qci3	0
saegw-pgw-bearers-setup-success-qci4	0
saegw-pgw-bearers-setup-success-qci5	0
saegw-pgw-bearers-setup-success-qci6	0
saegw-pgw-bearers-setup-success-qci7	0
saegw-pgw-bearers-setup-success-qci8	0
saegw-pgw-bearers-setup-success-qci9	60
saegw-pgw-bearers-setup-success-qci-other	0
saegw-pgw-bearers-setup-fail-qci1	0
saegw-pgw-bearers-setup-fail-qci2	0
saegw-pgw-bearers-setup-fail-qci3	0
saegw-pgw-bearers-setup-fail-qci4	0
saegw-pgw-bearers-setup-fail-qci5	0
saegw-pgw-bearers-setup-fail-qci6	0
saegw-pgw-bearers-setup-fail-qci7	0
saegw-pgw-bearers-setup-fail-qci8	0
saegw-pgw-bearers-setup-fail-qci9	0
saegw-pgw-bearers-setup-fail-qci-other	0
local-close-attempts-await-coa-timeout	0
l2tp-gateway-session-delete-attempts	0
uplink-packets	77233
uplink-bytes	7799143
uplink-drop-packets	1258
uplink-drop-bytes	181524
downlink-packets	144952
downlink-bytes	181906130
downlink-drop-packets	294
downlink-drop-bytes	404774
uplink-suspend-drop-packets	0
uplink-suspend-drop-bytes	0
downlink-suspend-drop-packets	0
downlink-suspend-drop-bytes	0
num-suspend-notification	0
num-suspend-notification-accepts	0
num-suspend-notification-rejects	0
num-resume-notification	0
num-resume-notification-accepts	0
num-resume-notification-rejects	0
delete-pdp-context-drops-teardown-ind-missing-single-bearer-session	0
delete-pdp-context-drops-teardown-ind-missing-default-bearer	0
s6b-initiated-close-attempts	0
default-gy-uplink-bytes	0

default-gy-downlink-bytes	0
num-active-s6b-sessions	0
number-of-sessions-ignored-radius-failure	0
number-of-inter-access-handover-req-no-session-rejects	0
number-of-inter-access-handover-req-no-session-accepts	0
num-dns-cache-hits	0
num-dns-cache-hits-ntp	0
num-dns-cache-hits-srv	0
num-dns-cache-hits-ip	0
num-dns-attempts	0
num-dns-send-success	0
num-dns-send-failed-no-config	0
num-dns-send-failed-misc	0
num-dns-responses	0
num-dns-response-success	0
num-dns-response-failed-no-response	0
num-dns-response-failed-name-not-found	0
num-dns-response-failed-no-data	0
num-dns-response-failed-misc	0
num-session-dns-timeout	0
num-dns-response-dropped-no-interface	0
num-dns-response-dropped-no-session	0
num-dns-response-dropped-no-pdn-session	0
num-session-pended-for-dns	0
num-session-pended-dns-success	0
num-session-pended-dns-timeout	0
pco-pcscf-default-fqdn-used	0
pco-pcscf-table-used	0
num-saegw-pgw-uplink-packets	77233
num-saegw-pgw-uplink-bytes	7799143
num-saegw-pgw-downlink-packets	144952
num-saegw-pgw-downlink-bytes	181906130
num-saegw-pgw-uplink-drop-packets	1258
num-saegw-pgw-uplink-drop-bytes	181524
num-saegw-pgw-downlink-drop-packets	294
num-saegw-pgw-downlink-drop-bytes	404774
num-uplink-s5s8-geran-packets	0
num-uplink-s5s8-geran-bytes	0
num-downlink-s5s8-geran-packets	0
num-downlink-s5s8-geran-bytes	0
num-uplink-s5s8-geran-drop-packets	0
num-uplink-s5s8-geran-drop-bytes	0
num-downlink-s5s8-geran-drop-packets	0
num-downlink-s5s8-geran-drop-bytes	0
num-saegw-pgw-uplink-s5s8-geran-packets	0
num-saegw-pgw-uplink-s5s8-geran-bytes	0
num-saegw-pgw-downlink-s5s8-geran-packets	0

num-saegw-pgw-downlink-s5s8-geran-bytes	0
num-saegw-pgw-uplink-s5s8-geran-drop-packets	0
num-saegw-pgw-uplink-s5s8-geran-drop-bytes	0
num-saegw-pgw-downlink-s5s8-geran-drop-packets	0
num-saegw-pgw-downlink-s5s8-geran-drop-bytes	0
num-uplink-s5s8-utran-packets	0
num-uplink-s5s8-utran-bytes	0
num-downlink-s5s8-utran-packets	0
num-downlink-s5s8-utran-bytes	0
num-uplink-s5s8-utran-drop-packets	0
num-uplink-s5s8-utran-drop-bytes	0
num-downlink-s5s8-utran-drop-packets	0
num-downlink-s5s8-utran-drop-bytes	0
num-saegw-pgw-uplink-s5s8-utran-packets	0
num-saegw-pgw-uplink-s5s8-utran-bytes	0
num-saegw-pgw-downlink-s5s8-utran-packets	0
num-saegw-pgw-downlink-s5s8-utran-bytes	0
num-saegw-pgw-uplink-s5s8-utran-drop-packets	0
num-saegw-pgw-uplink-s5s8-utran-drop-bytes	0
num-saegw-pgw-downlink-s5s8-utran-drop-packets	0
num-saegw-pgw-downlink-s5s8-utran-drop-bytes	0
num-uplink-eutran-packets	77233
num-uplink-eutran-bytes	7799143
num-downlink-eutran-packets	144952
num-downlink-eutran-bytes	181906130
num-uplink-eutran-drop-packets	1258
num-uplink-eutran-drop-bytes	181524
num-downlink-eutran-drop-packets	294
num-downlink-eutran-drop-bytes	404774
num-saegw-pgw-uplink-eutran-packets	77233
num-saegw-pgw-uplink-eutran-bytes	7799143
num-saegw-pgw-downlink-eutran-packets	144952
num-saegw-pgw-downlink-eutran-bytes	181906130
num-saegw-pgw-uplink-eutran-drop-packets	1258
num-saegw-pgw-uplink-eutran-drop-bytes	181524
num-saegw-pgw-downlink-eutran-drop-packets	294
num-saegw-pgw-downlink-eutran-drop-bytes	404774
num-uplink-nbiot-eutran-packets	0
num-uplink-nbiot-eutran-bytes	0
num-downlink-nbiot-eutran-packets	0
num-downlink-nbiot-eutran-bytes	0
num-uplink-nbiot-eutran-drop-packets	0
num-uplink-nbiot-eutran-drop-bytes	0
num-downlink-nbiot-eutran-drop-packets	0
num-downlink-nbiot-eutran-drop-bytes	0
num-saegw-pgw-uplink-nbiot-eutran-packets	0
num-saegw-pgw-uplink-nbiot-eutran-bytes	0

num-saegw-pgw-downlink-nbiot-eutran-packets	0
num-saegw-pgw-downlink-nbiot-eutran-bytes	0
num-saegw-pgw-uplink-nbiot-eutran-drop-packets	0
num-saegw-pgw-uplink-nbiot-eutran-drop-bytes	0
num-saegw-pgw-downlink-nbiot-eutran-drop-packets	0
num-saegw-pgw-downlink-nbiot-eutran-drop-bytes	0
uplink-gngp-geran-packets	0
uplink-gngp-geran-bytes	0
uplink-gngp-geran-drop-packets	0
uplink-gngp-geran-drop-bytes	0
downlink-gngp-geran-packets	0
downlink-gngp-geran-bytes	0
downlink-gngp-geran-drop-packets	0
downlink-gngp-geran-drop-bytes	0
uplink-gngp-utran-packets	0
uplink-gngp-utran-bytes	0
uplink-gngp-utran-drop-packets	0
uplink-gngp-utran-drop-bytes	0
downlink-gngp-utran-packets	0
downlink-gngp-utran-bytes	0
downlink-gngp-utran-drop-packets	0
downlink-gngp-utran-drop-bytes	0
num-spr-auth-requests	0
num-spr-auth-req-accepts	0
num-spr-auth-req-rejects	0
num-spr-auth-req-timeouts	0
num-acct-start-requests	0
num-acct-start-req-accepts	0
num-acct-start-req-timeouts	0
num-acct-start-req-not-waiting-for-response	0
num-acct-interim-update-requests	0
num-acct-interim-update-req-accepts	0
num-acct-interim-update-req-timeouts	0
num-acct-interim-update-req-not-waiting-for-response	0
num-acct-stop-requests	0
num-acct-stop-req-accepts	0
num-acct-stop-req-timeouts	0
num-acct-stop-req-not-waiting-for-response	0
num-ttl-excd-drop-packets	606
num-ttl-excd-drop-bytes	89256
num-frag-send-packets	0
num-frag-send-bytes	0
num-frag-fail-packets	0
num-frag-fail-bytes	0
num-uplink-acl-drop-packets	0
num-uplink-acl-drop-bytes	0
num-downlink-acl-drop-packets	0

num-downlink-acl-drop-bytes	0
num-bad-src-addr-drop-packets	0
num-bad-src-addr-drop-bytes	0
num-no-tft-drop-packets	0
num-no-tft-drop-bytes	0
num-bad-ue-addr-drop-packets	0
num-bad-ue-addr-drop-bytes	0
num-mtu-drop-packets	0
num-mtu-drop-bytes	0
num-no-route-drop-packets	0
num-no-route-drop-bytes	0
num-no-egress-intf-drop-packets	0
num-no-egress-intf-drop-bytes	0
num-cdrs-with-traffic-and-no-sr-stats	0
num-session-start-pilot-packets-sent	0
num-session-stop-pilot-packets-sent	0
num-rat-change-pilot-packets-sent	0
create-session-accepts-apn-matching-type-exact-match	80
create-session-accepts-apn-matching-type-ni-only	0
create-session-accepts-apn-matching-type-ni-and-malformed-oi	0
create-pdp-session-accepts-apn-matching-type-exact-match	0
create-pdp-session-accepts-apn-matching-type-ni-only	0
create-pdp-session-accepts-apn-matching-type-ni-and-malformed-oi	0
delete-session-rejects-mismatched-src-and-ftaid-ip	0
delete-pdp-context-rejects-mismatched-src-and-ftaid-ip	0
modify-bearer-rejects-mismatched-src-and-ftaid-ip	0
update-pdp-context-rejects-mismatched-src-and-ftaid-ip	0
local-close-parallel-pdnconnection-setup	0
local-close-parallel-pdp-context-setup	0
delete-session-req-rejects-mismatched-peer-ip-or-imsi	0
delete-pdp-context-req-rejects-mismatched-peer-ip-or-imsi	0
create-bearer-resp-drops-mismatched-peer-ip-or-imsi	0
bearer-resource-command-rejects-mismatched-peer-ip-or-imsi	0
create-pdp-context-req-rejects-mismatched-peer-ip-or-imsi	0
change-notification-req-rejects-mismatched-peer-ip-or-imsi	0
modify-bearer-req-rejects-mismatched-peer-ip-or-imsi	0
update-pdp-context-req-rejects-mismatched-peer-ip-or-imsi	0
delete-bearer-command-rejects-mismatched-peer-ip-or-imsi	0
delete-bearer-resp-drops-mismatched-peer-ip-or-imsi	0
delete-pdp-context-resp-drops-mismatched-peer-ip-or-imsi	0
modify-bearer-command-rejects-mismatched-peer-ip-or-imsi	0
update-bearer-resp-drops-mismatched-peer-ip-or-imsi	0
update-pdp-contexts-resp-drops-mismatched-peer-ip-or-imsi	0
init-pdp-context-activation-resp-drops-mismatched-peer-ip-or-imsi	0
trace-session-activation-req-rejects-mismatched-peer-ip-or-imsi	0
trace-session-deactivation-req-rejects-mismatched-peer-ip-or-imsi	0
suspend-notification-req-rejects-mismatched-peer-ip-or-imsi	0

resume-notification-req-rejects-mismatched-peer-ip-or-imsi	0
minimum-control-plane-teid-reuse-duration	4294967295
minimum-user-plane-teid-reuse-duration	4294967295
num-stats-update-for-missing-srs-for-active-sessions	0
num-stats-update-for-missing-srs-for-deactivate-sessions	0
num-cdrs-with-zero-service-data-containers	0
num-sig-info-lookup-successful-and-filled-configured-values	0
num-sig-info-lookup-failure	0
number-of-times-ocs-down-session-terminate-timer-started	0
number-of-active-sessions-with-ocs-down-session-terminate-timer-running	0
create-session-rejects-session-blocked	0
local-close-attempts-session-blocked	0
local-close-terminate-rule	0
create-session-rejects-tam-failure	0
create-pdp-context-rejects-tam-failure	0
create-session-reject-proc-limit	0
create-pdp-reject-proc-limit	0
delete-session-drop-proc-limit	0
delete-pdp-drop-proc-limit	0
num-sessios-in-local-database	0
sess-setup-failures-external-errors	0
sess-deactivations-external-errors	0
sess-setup-failures-local-errors	0
sess-deactivations-local-errors	20
ConnectedDevice-flows	0
ConnectedDevice-uplink-packets	0
ConnectedDevice-uplink-bytes	0
ConnectedDevice-downlink-packets	0
ConnectedDevice-downlink-bytes	0
TetheredDevice-flows	0
TetheredDevice-uplink-packets	0
TetheredDevice-uplink-bytes	0
TetheredDevice-downlink-packets	0
TetheredDevice-downlink-bytes	0
create-cups-session-attempts	0
create-cups-session-success	0
create-cups-session-failure	0
modify-cups-session-attempts	0
modify-cups-session-success	0
modify-cups-session-failure	0
delete-cups-session-attempts	0
delete-cups-session-success	0
delete-cups-session-failure	0
num-local-close-session-monitor	0
current-cups-sessions	0
TetheringEntitled-flows	0
TetheringEntitled-uplink-packets-passed	0

TetheringEntitled-uplink-bytes-passed	0
TetheringEntitled-downlink-packets-passed	0
TetheringEntitled-downlink-bytes-passed	0
TetheredFlowsDetected-ttl	0
TetheredFlowsDetected-user-agent	0
num-pgw-triggered-qos-update	0
create-session-accepts-multi-pdnconnections-same-apn	0
TetheredFlowsDetected-tcp-ip-fingerprinting	0
TetheredFlowsDetected-gquic-user-agent	0
3gpp-to-non-3gpp-handover-delayed	0
3gpp-to-non-3gpp-delayed-handover-completed-on-first-ulPkt	0
3gpp-to-non-3gpp-delayed-handover-completed-on-timer-expiry	0
3gpp-to-non-3gpp-delayed-handover-completed-on-b2b-handover	0
3gpp-to-non-3gpp-delayed-handover-completed-on-other-event	0
number-of-appl-start-notifications-rcvd	0
number-of-appl-stop-notifications-rcvd	0
number-of-gx-appl-start-notifications-success	0
number-of-gx-appl-start-notifications-failed	0
number-of-gx-appl-stop-notifications-success	0
number-of-gx-appl-stop-notifications-failed	0
num-session-monitor-modifications-sent	0
number-of-sd-appl-start-notifications-success	0
number-of-sd-appl-start-notifications-failed	0
number-of-sd-appl-stop-notifications-success	0
number-of-sd-appl-stop-notifications-failed	0
num-local-close-pcrf-down-session-hold-time	0
num-create-session-rejects-overload	0
num-create-bearer-rejects-overload	0
create-session-rejects-accounting-on-pending	0
create-pdp-contexts-rejects-accounting-on-pending	0
num-manifest-server-requests-sent	0
num-manifest-server-responses-rcvd	0
TetheredFlowsReported-ttl	0
TetheredFlowsReported-user-agent	0
TetheredFlowsReported-tcp-ip-fingerprinting	0
TetheredFlowsReported-gquic-user-agent	0
num-create-session-rejects-overload-minor	0
num-create-session-rejects-overload-major	0
num-create-session-rejects-overload-critical	0
num-create-session-rejects-overload-standby-sync	0
num-create-bearer-rejects-overload-minor	0
num-create-bearer-rejects-overload-major	0
num-create-bearer-rejects-overload-critical	0
num-create-bearer-rejects-overload-standby-sync	0
discontinuity-timestamp	1579635095

Peer Chassis-level GTP Statistics Table

Use the **gtp peer** parameter to display GTP performance statistics by peer.

```
an3000-mcm-slot7cpu1# show zone default gateway statistics
grid-level gtp peer <peer-name>
```

```
gateway statistics grid-level gtp peer GN-sgw-10.64.133.32
create-session-request-decode-success 0
create-session-request-encode-success 80
create-session-request-decode-failure 0
create-session-request-encode-failure 0
create-session-request-rcvd-retx 0
create-session-response-decode-success 80
create-session-response-encode-success 0
create-session-response-decode-failure 0
create-session-response-encode-failure 0
delete-session-request-decode-success 0
delete-session-request-encode-success 1
delete-session-request-decode-failure 0
delete-session-request-encode-failure 0
delete-session-response-decode-success 1
delete-session-response-encode-success 0
delete-session-response-decode-failure 0
delete-session-response-encode-failure 0
modify-bearer-request-decode-success 0
modify-bearer-request-encode-success 79
modify-bearer-request-decode-failure 0
modify-bearer-request-encode-failure 0
modify-bearer-response-decode-success 79
modify-bearer-response-encode-success 0
modify-bearer-response-decode-failure 0
modify-bearer-response-encode-failure 0
create-bearer-request-decode-success 0
create-bearer-request-encode-success 0
create-bearer-request-decode-failure 0
create-bearer-request-encode-failure 0
create-bearer-response-decode-success 0
create-bearer-response-encode-success 0
create-bearer-response-decode-failure 0
create-bearer-response-encode-failure 0
delete-bearer-request-decode-success 77
delete-bearer-request-encode-success 0
delete-bearer-request-decode-failure 0
delete-bearer-request-encode-failure 0
delete-bearer-response-decode-success 0
delete-bearer-response-encode-success 77
delete-bearer-response-decode-failure 0
```

delete-bearer-response-encode-failure	0
bearer-resource-command-decode-success	0
bearer-resource-command-encode-success	0
bearer-resource-command-decode-failure	0
bearer-resource-command-encode-failure	0
bearer-resource-failure-indication-decode-success	0
bearer-resource-failure-indication-encode-success	0
bearer-resource-failure-indication-decode-failure	0
bearer-resource-failure-indication-encode-failure	0
modify-bearer-command-decode-success	0
modify-bearer-command-encode-success	0
modify-bearer-command-decode-failure	0
modify-bearer-command-encode-failure	0
modify-bearer-failure-indication-decode-success	0
modify-bearer-failure-indication-encode-success	0
modify-bearer-failure-indication-decode-failure	0
modify-bearer-failure-indication-encode-failure	0
update-bearer-request-decode-success	0
update-bearer-request-encode-success	0
update-bearer-request-decode-failure	0
update-bearer-request-encode-failure	0
update-bearer-response-decode-success	0
update-bearer-response-encode-success	0
update-bearer-response-decode-failure	0
update-bearer-response-encode-failure	0
downlink-data-notification-decode-success	0
downlink-data-notification-encode-success	0
downlink-data-notification-decode-failure	0
downlink-data-notification-encode-failure	0
downlink-data-notification-ack-decode-success	0
downlink-data-notification-ack-encode-success	0
downlink-data-notification-ack-decode-failure	0
downlink-data-notification-ack-encode-failure	0
downlink-data-notification-failure-indication-decode-success	0
downlink-data-notification-failure-indication-encode-success	0
downlink-data-notification-failure-indication-decode-failure	0
downlink-data-notification-failure-indication-encode-failure	0
release-access-bearers-request-decode-success	0
release-access-bearers-request-encode-success	0
release-access-bearers-request-decode-failure	0
release-access-bearers-request-encode-failure	0
release-access-bearers-response-decode-success	0
release-access-bearers-response-encode-success	0
release-access-bearers-response-decode-failure	0
release-access-bearers-response-encode-failure	0
change-notification-request-decode-success	0
change-notification-request-encode-success	0

change-notification-request-decode-failure	0
change-notification-request-encode-failure	0
change-notification-response-decode-success	0
change-notification-response-encode-success	0
change-notification-response-decode-failure	0
change-notification-response-encode-failure	0
create-pdp-context-request-decode-success	0
create-pdp-context-request-encode-success	0
create-pdp-context-request-decode-failure	0
create-pdp-context-request-encode-failure	0
create-pdp-context-request-rcvd-retx	0
create-pdp-context-response-decode-success	0
create-pdp-context-response-encode-success	0
create-pdp-context-response-decode-failure	0
create-pdp-context-response-encode-failure	0
delete-pdp-context-request-decode-success	0
delete-pdp-context-request-encode-success	0
delete-pdp-context-request-decode-failure	0
delete-pdp-context-request-encode-failure	0
delete-pdp-context-response-decode-success	0
delete-pdp-context-response-encode-success	0
delete-pdp-context-response-decode-failure	0
delete-pdp-context-response-encode-failure	0
update-pdp-context-request-decode-success	0
update-pdp-context-request-encode-success	0
update-pdp-context-request-decode-failure	0
update-pdp-context-request-encode-failure	0
update-pdp-context-response-decode-success	0
update-pdp-context-response-encode-success	0
update-pdp-context-response-decode-failure	0
update-pdp-context-response-encode-failure	0
num-echo-requests-received	0
num-echo-request-decode-failure	0
num-echo-requests-sent	0
num-echo-responses-received	0
num-echo-response-decode-failure	0
num-echo-responses-sent	0
num-trace-session-activation-decode-success	0
num-trace-session-activation-encode-success	0
num-trace-session-activation-decode-failure	0
num-trace-session-activation-encode-failure	0
num-trace-session-deactivation-decode-success	0
num-trace-session-deactivation-encode-success	0
num-trace-session-deactivation-decode-failure	0
num-trace-session-deactivation-encode-failure	0
num-create-indirect-data-forwarding-tunnel-request-decode-success	0
num-create-indirect-data-forwarding-tunnel-request-decode-failure	0

```

num-create-indirect-data-forwarding-tunnel-response-encode-success 0
num-delete-indirect-data-forwarding-tunnel-request-decode-success 0
num-delete-indirect-data-forwarding-tunnel-request-decode-failure 0
num-delete-indirect-data-forwarding-tunnel-response-encode-success 0
num-suspend-notification-decode-success 0
num-suspend-notification-encode-success 0
num-suspend-notification-decode-failure 0
num-suspend-notification-encode-failure 0
num-suspend-acknowledge-decode-success 0
num-suspend-acknowledge-encode-success 0
num-suspend-acknowledge-decode-failure 0
num-suspend-acknowledge-encode-failure 0
num-resume-notification-decode-success 0
num-resume-notification-encode-success 0
num-resume-notification-decode-failure 0
num-resume-notification-encode-failure 0
num-resume-acknowledge-decode-success 0
num-resume-acknowledge-encode-success 0
num-resume-acknowledge-decode-failure 0
num-resume-acknowledge-encode-failure 0
create-session-response-with-piggyback-decode-success 0
create-session-response-with-piggyback-decode-failure 0
num-sxab-sess-estab-req-decode-success 0
num-sxab-sess-estab-req-encode-success 0
num-sxab-sess-estab-req-decode-failure 0
num-sxab-sess-estab-req-encode-failure 0
num-sxab-sess-estab-resp-decode-success 0
num-sxab-sess-estab-resp-encode-success 0
num-sxab-sess-estab-resp-decode-failure 0
num-sxab-sess-estab-resp-encode-failure 0
num-sxab-sess-mod-req-decode-success 0
num-sxab-sess-mod-req-encode-success 0
num-sxab-sess-mod-req-decode-failure 0
num-sxab-sess-mod-req-encode-failure 0
num-sxab-sess-mod-resp-decode-success 0
num-sxab-sess-mod-resp-encode-success 0
num-sxab-sess-mod-resp-decode-failure 0
num-sxab-sess-mod-resp-encode-failure 0
num-sxab-sess-term-req-decode-success 0
num-sxab-sess-term-req-encode-success 0
num-sxab-sess-term-req-decode-failure 0
num-sxab-sess-term-req-encode-failure 0
num-sxab-sess-term-resp-decode-success 0
num-sxab-sess-term-resp-encode-success 0
num-sxab-sess-term-resp-decode-failure 0
num-sxab-sess-term-resp-encode-failure 0
num-sxab-report-decode-success 0

```

```

num-sxab-report-encode-success 0
num-sxab-report-decode-failure 0
num-sxab-report-encode-failure 0
num-sxab-report-ack-decode-success 0
num-sxab-report-ack-encode-success 0
num-sxab-report-ack-decode-failure 0
num-sxab-report-ack-encode-failure 0
num-sxab-sess-status-req-decode-success 0
num-sxab-sess-status-req-encode-success 0
num-sxab-sess-status-req-decode-failure 0
num-sxab-sess-status-req-encode-failure 0
num-sxab-sess-status-resp-decode-success 0
num-sxab-sess-status-resp-encode-success 0
num-sxab-sess-status-resp-decode-failure 0
num-sxab-sess-status-resp-encode-failure 0
discontinuity-timestamp
1579635204

```

IMSI Manager Message Statistics

The IMSI manager task runs on the CSM and maintains a database of all IMSIs registered on the MCC. It streamlines handling of GTP-C packets to ensure the SGW/PGW/GGSN task remains the same during a session's lifetime. It also pulls load information for each task to evenly distribute session requests across the tasks.

Use the following command to display the IMSI manager task (IMSIMGR) statistics:

```

an3000-mcm-slot7cpu1# show zone default gateway debug-stats
imsimgr message-statistics

```

```

gateway debug-stats imsimgr message-statistics 32
number-of-gtpc-messages-received 80
number-of-gtpc-messages-dropped-not-synced 0
number-of-messages-dropped-no-gw 0
number-of-gtpc-messages-dropped-wrong-message-type 0
number-of-gtpc-messages-dropped-decode-fail 0
number-of-gtpc-messages-identified-by-sequence-number 0
number-of-dhcp-messages-received 0
number-of-dhcp-message-received-no-existing-subscriber 0
number-of-dhcp-messages-dropped-not-synced 0
number-of-dhcp-messages-dropped-no-imsi 0
number-of-dhcp-messages-forwarded 0
number-of-ip-packets-received 0
number-of-ip-packets-dropped-not-synced 0
number-of-ip-packets-dropped-no-imsi 0
number-of-ip-packets-dropped-no-subscriber 0
number-of-gtpu-messages-received 1
number-of-gtpu-messages-dropped-not-synced 0

```

number-of-gtpu-messages-dropped-unknown-type	0
number-of-gtpu-messages-dropped-decode-failure	0
number-of-gtpu-error-indications-received	1
number-of-radius-messages-received	0
number-of-radius-messages-dropped-not-synced	0
number-of-radius-messages-dropped-decode-fail	0
number-of-security-policy-manager-messages-received	0
number-of-security-policy-manager-messages-received-with-imsi	0
number-of-security-policy-manager-messages-dropped-not-synced	0
number-of-security-policy-manager-messages-dropped-no-identity	0
number-of-security-policy-manager-messages-dropped-no-pseudonym-identity	0
number-create-request-forwarded-no-peer-pgw	0
number-create-request-dropped-previous-gw-overload	0
number-create-request-dropped-send-batch-fail	0
number-create-request-dropped-add-to-batch-fail	0
number-change-notification-request-dropped-add-to-batch-fail	0
number-change-notification-response-dropped-add-to-batch-fail	0
number-change-notification-request-dropped-decode-fail	0
number-change-notification-response-dropped-decode-fail	0
number-create-request-dropped-slb-tbl-empty	0
number-create-request-dropped-sgw-cant-accept	0
number-create-request-dropped-pgw-cant-accept	0
number-create-request-added-to-batch	80
number-change-notification-request-pinned-sgw-rate-ctrl-dropped	0
number-change-notification-response-pinned-sgw-rate-ctrl-dropped	0
number-change-notification-request-dropped-no-gw	0
number-change-notification-response-dropped-no-gw	0
number-change-notification-request-dropped-sgw-cant-accept	0
number-change-notification-response-dropped-sgw-cant-accept	0
number-change-notification-request-dropped-pgw-cant-accept	0
number-change-notification-response-dropped-pgw-cant-accept	0
number-change-notification-request-pinned-pgw-rate-ctrl-dropped	0
number-change-notification-response-pinned-pgw-rate-ctrl-dropped	0
number-create-requests-forwarded	80
number-create-request-send-batch-successes	80
number-create-request-send-batch-failures	0
number-create-request-resend-batch-success	0
number-create-request-resend-batch-failure	0
number-create-request-unmatched-new-peer-pgw	0
number-create-request-unmatched-peer-sgw	0
number-create-request-unmatched-peer-pgw	0
number-create-request-resend-failure-no-gw	0
number-create-request-pinned-sgw-rate-ctrl-dropped	0
number-create-request-pinned-pgw-rate-ctrl-dropped	0
number-emergency-create-request-received	0
number-emergency-create-request-dropped-add-to-batch-fail	0
number-sgw-imsi-in-gw-list	79

number-pgw-imsi-in-gw-list	0
number-sgw-imsi-new-gw-selected	1
number-pgw-imsi-new-gw-selected	0
number-pinned-sgw-selected-pgw	0
number-pinned-sgw-pinned-pgw	79
number-selected-sgw-pinned-pgw	0
number-selected-sgw-selected-pgw	1
number-of-diameter-request-to-send-to-preferred-gw	0
number-of-diameter-request-received	0
number-of-diameter-request-dropped-decode-fail	0
number-of-diameter-request-dropped-invalid-format	0
number-of-diameter-request-dropped-invalid-key	0
number-of-diameter-request-dropped-no-imsi	0
number-of-diameter-request-dropped-subscriber-not-found	0
number-of-diameter-request-dropped-task-not-found	0
number-of-diameter-request-sent-to-gw	0
number-of-diameter-request-add-to-batch-fail	0
number-create-request-rejected-admission-control	0
number-of-session-learner-messages-dropped-not-synced	0
number-of-session-learner-manager-messages-received	0
number-of-session-learner-messages-sent-to-gw	0
number-of-session-learner-messages-add-to-batch-fail	0
number-radius-username-anonymous-get-identifier-from-nai-failed	0
non-create-messages-added-to-batch-fail-for-gateway-agent	0
non-create-messages-added-to-batch-fail-for-pdn-gateway	0
non-create-messages-added-to-batch-fail-for-s-gateway	0
number-create-requests-added-to-batch-for-gateway-agent	0
number-create-requests-added-to-batch-gw-for-pdn-gateway	0
number-create-requests-added-to-batch-for-sgw	0
number-create-requests-forwarded-for-gateway-agent	0
number-create-requests-forwarded-for-pdn-gateway	0
number-create-requests-forwarded-for-sgw	0
number-non-create-messages-dropped-added-to-batch-fail	0
number-pfcp-create-messages-received	0
number-pfcp-messages-dropped-added-to-batch-fail	0
number-pfcp-messages-dropped-decode-failures	0
number-pfcp-messages-dropped-no-seid	0
number-pfcp-messages-dropped-wrong-message-type	0
number-pfcp-messages-dropped-not-synced	0
number-pfcp-messages-received	0
number-valid-non-create-pfcp-messages-received	0

GTP Peer Status

Use the **gtp-peer-status** parameter to display a status update on the GTP peering between the MCC gateway (SGW/PGW) and its GTP peer.

```
an3000-mcm-slot7cpu1# show zone default gateway gtp-peer-status
```

```
gateway gtp-peer-status
```

GATEWAY NAME	PEER IP ADDRESS	NETWORK CONTEXT	LOCAL IP ADDRESS	GTP VERSION	PEER	PEER	PEER	PEER STATE
					RESTART COUNTER	NODE LCI	NODE OCI	
10.64.140.6-2152	10.71.32.12		10.64.140.6	gtpUv1	0	0	0	unknown
NYsgw-1	10.71.32.12	NC-GN-1	10.64.128.6	gtpv2	0	0	0	up

GTP Peer Status Filtering by Network Context

Prior to release 9.2, you can use the following command to filter GTP peer status by network-context. This command only displays the GTP peers that are configured on the specified network-context, and filters out all other network contexts.

```
an3000-mcm-slot7cpu1# show zone default gateway gtp-peer-status <network-context>
```

From release 9.2, you cannot use the command above to filter by network-context. You must use the following command and include the gateway name and gateway peer IP in addition to the network-context.

```
an3000-mcm-slot7cpu1# show zone default gateway gtp-peer-status <gateway-name> <peer-ip> <network-context>
```

Service Status Tables

Use the **service-status** parameter to display a snapshot of the health and utilization of the gateway.

```
an3000-mcm-slot7cpu1# show zone default gateway service-status ?
```

Possible completions:

```
epdg      ePDG Service Status Table
ggsn      GGSN Service Status Table
gtpproxy  GTP_PROXY Service Status Table
mcc-gigw  MCC Service Status Table
pgw       PGW Service Status Table
sgw       SGW Service Status Table
userplane User Plane Service Status Table
wag       WAG Service Status Table
|         Output modifiers
<cr>
```

For example:

```
an3000-mcm-slot7cpu1# show zone default gateway service-status sgw
```

```
gateway service-status sgw NYsgw-1
  geo-redundant-group      ""
  active-operational-status up
  number-of-active-instances 8
  standby-operational-status up
  number-of-standby-instances 8
  number-of-sessions        1
  number-of-PDN-sessions    1
  number-of-active-pdn-sessions 1
```

SLB Load Status Table

Use the **slb-load-status** parameter to display the load distribution status across the gateways.

```
an3000-mcm-slot7cpu1# show zone default gateway slb-load-
status
```

```
gateway slb-load-status
```

GATEWAY			LOAD	CURRENT	ABSOLUTE
TYPE	NAME	SERVICEID	STATE	LOAD	VALUE

sgw	SGW_6_9_1_0_SGW_0	85	MinorLow	0	0
sgw	SGW_6_9_1_1_SGW_2	88	MinorLow	0	0
sgw	SGW_6_9_1_2_SGW_4	100	MinorLow	0	0
sgw	SGW_6_9_1_3_SGW_6	112	MinorLow	0	1
sgw	SGW_6_9_1_4_SGW_8	115	MinorLow	0	0
sgw	SGW_6_9_1_5_SGW_10	118	MinorLow	0	0
sgw	SGW_6_9_1_6_SGW_12	121	MinorLow	0	0
sgw	SGW_6_9_1_7_SGW_14	124	MinorLow	0	0
pgw	PDNGW_6_9_1_0_PDNGW_0	37	MinorLow	0	0
pgw	PDNGW_6_9_1_1_PDNGW_2	40	MinorLow	0	0
pgw	PDNGW_6_9_1_2_PDNGW_4	43	MinorLow	0	0
pgw	PDNGW_6_9_1_3_PDNGW_6	46	MinorLow	0	1
pgw	PDNGW_6_9_1_4_PDNGW_8	49	MinorLow	0	0
pgw	PDNGW_6_9_1_5_PDNGW_10	52	MinorLow	0	0
pgw	PDNGW_6_9_1_6_PDNGW_12	55	MinorLow	0	0
pgw	PDNGW_6_9_1_7_PDNGW_14	58	MinorLow	0	0
u-plane	GwAgent_6_9_1_0_GwAgent_0	61	MinorLow	0	0
u-plane	GwAgent_6_9_1_1_GwAgent_2	64	MinorLow	0	0
u-plane	GwAgent_6_9_1_2_GwAgent_4	67	MinorLow	0	1
u-plane	GwAgent_6_9_1_3_GwAgent_6	70	MinorLow	0	0
u-plane	GwAgent_6_9_1_4_GwAgent_8	73	MinorLow	0	0
u-plane	GwAgent_6_9_1_5_GwAgent_10	76	MinorLow	0	0
u-plane	GwAgent_6_9_1_6_GwAgent_12	79	MinorLow	0	0
u-plane	GwAgent_6_9_1_7_GwAgent_14	82	MinorLow	0	0

Network Context Commands

Networking of the MCC is based on the concept of a network context, used by Virtual Routing and Forwarding (VRF), which consists of multiple, independent routing instances/tables that are created on the MCC.

Table 1-3 MCC Network Contexts

Network Context Name	Description
GnGp-NC ^a	Interface to SGSN
Gx-NC	Interface to PCRF
Gy-NC	Interface to OCS
GiSGi-NC ^a	Interface to Internet
management-context	Supports OAM traffic, including EMS

^a Bearer network contexts

show network context GnGp-NC ip interface

```
an3000-mcm-slot7cpu1# show network-context GN ip interface
```

```
ip interface gn_interface
  IP Address          10.64.32.32
  IP Prefix Length    24
  IP Address          2001:470:8865:4020::20
  IP Prefix Length    64
  OperStatus          UP
  TxV4Pkts            923953
  TxV4Bytes           263454822
  TxV4DropPkts        0
  TxV4DropBytes       0
  TxKernelV4IcmpPkts  777714
  TxErrorIndicationPkts 0
  TxErrorIndicationFailurePkts 0
  TxErrorIndicationFabricPkts 0
  TxErrorIndicationRateLimitPkts 0
  TxV4CsmProxyPkts    0
  RxV4Pkts            4615831
  RxV4Bytes           571405377
  RxV4DropPkts        1
  RxV4DropBytes       60
  RxICMPv4Pkts        20408
  TxICMPv4Pkts        0
  RxV4UnSuppAddrPkts  0
  RxV4InvalidPkts     0
  RxV4MulticastPkts   0
```

RxV4CsmProxyPkts	0
RxV4UeUpstreamPkts	77879
RxV4UeDownstreamPkts	0
RxV4UeSubnetDownstreamPkts	0
RxV4LoadbalancerPkts	0
InvalidNatPortCounter	0
RxGtpInvalidPktscounter	0
TxV6Pkts	31
TxV6Bytes	2838
TxV6DropPkts	0
TxV6DropBytes	0
TxKernelV6IcmpPkts	31
TxV6CsmProxyPkts	0
RxV6Pkts	670917
RxV6Bytes	127670246
RxV6DropPkts	0
RxV6DropBytes	0
RxICMPv6Pkts	20894
RxV6UnSuppAddrPkts	0
RxV6InvalidPkts	0
RxV6IcmpNAPkts	0
RxV6IcmpTooBigPkts	0
RxV6CsmProxyPkts	0
RxV6UeUpstreamPkts	0
RxV6UeDownstreamPkts	0
RxV6UeSubnetDownstreamPkts	0
RxV6LoadbalancerPkts	0
TxICMPv6Pkts	0
AclDropPkts	0
AclDropBytes	0
AclAllowPkts	0
AclAllowBytes	0
AllTcpFlagsSetPktsDropped	0
AllTcpFlagsSetPktsDroppedIpv6	0
SynAckFinRstUrgPshFlagsSetPktsDropped	0
SynAckFinRstUrgPshFlagsSetPktsDroppedIpv6	0
SynRstAckUrgFlagsSetPktsDropped	0
SynRstAckUrgFlagsSetPktsDroppedIpv6	0
SynFinPshUrgFlagsSetPktsDropped	0
SynFinPshUrgFlagsSetPktsDroppedIpv6	0
FinPshUrgFlagsSetPktsDropped	0
FinPshUrgFlagsSetPktsDroppedIpv6	0
OnlyPshFlagSetPktsDropped	0
OnlyPshFlagSetPktsDroppedIpv6	0
OnlyFinFlagSetPktsDropped	0
OnlyFinFlagSetPktsDroppedIpv6	0
OnlyUrgFlagSetPktsDropped	0

OnlyUrgFlagSetPktsDroppedIpv6	0
FinRstFlagsSetPktsDropped	0
FinRstFlagsSetPktsDroppedIpv6	0
SynRstFlagsSetPktsDropped	0
SynRstFlagsSetPktsDroppedIpv6	0
FinSynFlagsSetPktsDropped	0
FinSynFlagsSetPktsDroppedIpv6	0
NoneTcpFlagsSetPktsDropped	0
NoneTcpFlagsSetPktsDroppedIpv6	0
TcpResrvedBitSetPktsDropped	0
TcpResrvedBitSetPktsDroppedIpv6	0
TcpSourcePortZeroPktsDropped	0
TcpSourcePortZeroPktsDroppedIpv6	0
TcpDestinationPortZeroPktsDropped	0
TcpDestinationPortZeroPktsDroppedIpv6	0
IncorrectTcpHeaderOffsetPktsDropped	0
IncorrectTcpHeaderOffsetPktsDroppedIpv6	0
IncorrectUdpHeadePktsDropped	0
IncorrectUdpHeadePktsDroppedIpv6	0
UdpSourcePortZeroPktsDropped	0
UdpSourcePortZeroPktsDroppedIpv6	0
UdpDestinationPortZeroPktsDropped	0
UdpDestinationPortZeroPktsDroppedIpv6	0
Ipv4HeaderErrorPktsDropped	0
Ipv4TruncatedPktsDropped	0
Ipv6HeaderErrorPktsDropped	0
MulticastIpv6AddrWithReservedValueZeroPktsDropped	0
MulticastIpv6AddrWithScopeFieldZeroPktsDropped	0
Ipv6AddrPayloadlengthZeroPktsDropped	0
Ipv6HopbyHopOptionEHParseFailPktsDropped	0
Ipv6HopbyHopOptionEHOrderFailPktsDropped	0
MulticastIpv6SrcAddrPktsDropped	0
Ipv6TruncatedPktsDropped	0
WrongIpVersionPktsDropped	0
InvalidPackets	0
InvalidTtlValuePktsDropped	0
Icmpv4HeaderErrorPktsDropped	0
Icmpv6HeaderErrorPackets	0
Icmpv6InvalidTypePackets	0
Icmpv6InvalidCodePackets	0
Icmpv4ParameterProblemTypePktsDropped	0
Icmpv4UnuseGwPktsDropped	0

You can apply additional parameters to the IP interface command to display specific fields. For example:

```
an3000-mcm-slot7cpu1# show network-context GN ip interface
oper-status
```

```
ip interface
OPER
IP IF NAME STATUS
-----
gn_interface UP
```

```
an3000-mcm-slot7cpu1# show network-context GN ip interface
rx-v4-bytes | select rx-v4-pkts | select tx-v4-bytes |
select tx-v4-pkts | select rx-v4-drop-bytes | select rx-v4-
drop-pkts | select tx-v4-drop-bytes | select tx-v4-drop-pkts
ip interface
```

IP IF NAME	TX				RX			
	TX V4		TX V4		RX V4		RX V4	
	PKTS	BYTES	DROP PKTS	DROP BYTES	PKTS	BYTES	DROP PKTS	DROP BYTES
gn_interface	928407	263826642	0	0	4646025	574455821	1	60

```
an3000-mcm-slot7cpu1# show network-context ip interface-
statistics
```

```
network-context GN
ip interface-statistics gn_interface
TxV4Pkts          930042
TxV4Bytes         263969004
TxV4DropPkts      0
TxV4DropBytes     0
RxV4Pkts          4657622
RxV4Bytes         575746647
RxV4DropPkts      1
RxV4DropBytes     60
RxICMPv4Pkts      20580
TxICMPv4Pkts      0
TxV6Pkts          31
TxV6Bytes         2838
TxV6DropPkts      0
TxV6DropBytes     0
TxICMPv6Pkts      0
RxV6Pkts          674864
RxV6Bytes         128182292
RxV6DropPkts      0
RxV6DropBytes     0
RxMplsPkts        0
RxMplsBytes       0
RxMplsUnknownLabelPkts 0
```

```

RxMplsDropPkts          0
TxMplsPkts              0
TxMplsBytes             0
RxICMPv6Pkts           0
TxNoRouteDropPkts      0
TxFragmentedPkts       0
Discontinuity Timestamp 1579635084

```

Proxy Commands

Use the commands and parameters in this section to view multi-protocol proxy cluster-level and node-level information and statistics.

Renamed HTTP Proxy Commands

In MCC release 8.4 and later, **http-proxy** was renamed to **multi-protocol-proxy** to encompass the following supported protocols: HTTP, TCP, HTTPS, UDP, and QUIC.

This change affects the following config and show commands:

- services http-proxy to services multi-protocol-proxy
- services crbn-profile <name> http-proxy to services crbn-profile <name> multi-protocol-proxy
- workflow data-profile <name> http-proxy to workflow data-profile <name> multi-protocol-proxy
- services http-proxy <name> http-ip-flow to services multi-protocol-proxy <name> ip-flow
- show services http-proxy to show services multi-protocol-proxy

multi-protocol-proxy connection

Use the following command to display information about the HTTP cluster proxy client and server connections:

```

an3000-mcm-slot7cpu1# show services multi-protocol-proxy
statistics cluster-level connection <chassis-id>

services multi-protocol-proxy statistics cluster-level
connection 32
Client Connections - Total                      4
Client Connections - Currently in Use          0
Client Connections - Keep-Alive Requested      2
Client Connections - Persistent                2
Client Connections - Reused                    2
Client Connections - Pipelined                 0
Client Connections - IPv4                      4
Client Connections - IPv6                      0

```

```
Server Connections - Total 2
Server Connections - Currently in Use 0
Server Connections - IPv4 2
Server Connections - IPv6 0
Server Connections - Currently Idle 0
Server Connections - Reused by Same Client 0
Server Connections - Reused from Pool 0
Server Connections - Preferred IP Family Used 0
Server Connections - Preferred IP Family Not Available 0
Smpp Connections - Total 0
Smpp Connections - Currently in Use 0
Smpp Connections - IPv4 0
Smpp Connections - IPv6 0
Smpp Connections - Currently Idle 0
Smpp Connections - Reused by Same Client 0
Smpp Connections - Reused 0
Smpp Connections -Connection Error 0
Smpp Connections - Bind Error 0
Smpp Connections - Send Error 0
Smpp Connections - Unbind Error 0
Server Connections - IP Transparent 0
Client Connections - Non-HTTP Message Received 0
Client Connections - Request Receive Timeouts 0
Client Connections - Reverse Handoffs 0
Client Connections - Failures 0
Fast Open - Attempts 0
Fast Open - Successes 0
Fast Open - Connection Initialization Failures 0
Fast Open - Connection Already in Pending List 0
Fast Open - Connection Already in Session 0
Fast Open - Policy Denied 0
Fast Open - Not Ready 0
Fast Open - Pending Connection List Full 0
Fast Open - Current Pending Connections 0
Fast Open - Pending Connection Idle Timeout 0
Discontinuity Timestamp
1579634984
```

multi-protocol-proxy summary

Proxy issues may be specific to a single ASM or a subset of the ASM VMs. Use the following command to check the health of each VM individually:

```
an3000-mcm-slot7cpu1# show services multi-protocol-proxy  
statistics node-level summary <chassis-id> <slot>
```

```

services multi-protocol-proxy statistics node-level summary
32 4
HTTP-Proxy Service - Attempts 0
HTTP-Proxy Service - Successes 0
HTTP-Proxy Service - Successfully Cached 0
HTTP-Proxy Service - Aborts 0
HTTP-Proxy Service - Failures 0
Client Requests - Total 0
Client Requests - Current 0
Client Requests - Parsed Successfully 0
Client Requests - Incomplete 0
Client Requests - Processing Failed 0
Client Requests - Pipeline Dropped 0
Client Requests - Invalid 0
Client Requests - Total Bytes 0
Client Responses - Total Bytes 0
Client Sessions - Current 0
Server Requests - Total Bytes 0
Server Responses - Total Bytes 0
Service Time - Min (ms) 0
Service Time - Avg (ms) 0
Service Time - Max (ms) 0
Server Responses - Total 0
Server Responses - Min Time (ms) 0
Server Responses - Avg Time (ms) 0
Server Responses - Max Time (ms) 0
Latency - Request Latency Average (ms) 0
Latency - Response Latency Average (ms) 0
Latency - Transaction Latency Average (ms) 0
Discontinuity Timestamp 1579634994

```

multi-protocol-proxy overload-control

Use the following command to check if the proxy has too much load:

```

an3000-mcm-slot7cpu1# show services multi-protocol-proxy
statistics cluster-level overload-control <chassis-id>

services multi-protocol-proxy statistics cluster-level
overload-control 32
Overload Control - Connection Accepts 877
Overload Control - Connection Denies 0
Discontinuity Timestamp 1579634984

```

IP Interface Packet Capture Utility

Use the following commands to perform a packet capture in pcapng format on a single IP interface configured in the system.

Note: This command affects system performance and should only be used for short durations when troubleshooting.

For example:

```
an3000-mcm-slot7cpu1# network-context

Possible completions:
  GI  GN  management-context

an3000-mcm-slot7cpu1# network-context GN ip-interface

Possible completions:
  gn_interface

an3000-mcm-slot7cpu1# network-context GN ip-interface gn_
interface

Possible completions:
  startcapture  Enable packet capture
  stopcapture   Disable packet capture

an3000-mcm-slot7cpu1# network-context GN ip-interface gn_
interface startcapture

Possible completions:
  count          max number of packets to capture, default is 1000
  duration        duration in seconds, default is 1
  file-name       file name prefix
  |              Output modifiers
```

In the example highlighted text, the packet trace is run against a single IP interface configured in a single network context. This means you can run a packet trace on either a Gn or Gi network context from a single command execution, but not both. However, you can start a packet capture on a Gn IP-interface and start another parallel packet capture on a Gi IP-interface during the same period, but not in the same single line command.

The packet captures are stored in `/var/log/eventlog.lnk` and can be retrieved using SCP/SFTP or EMS.

The filename format is `<filenameprefix><YYYYMMDDHHMMSS>.pcap`, for example, `PktTr_20140922111150.pcapng`. The file is saved as `pcapng` and is readable with Wireshark or similar packet analyzer software.

Packet Capture Considerations

In an MCC system with multiple active ingress/egress interfaces in each network context, traffic may be distributed among other interfaces, so the packet trace from a single interface may not represent the full traffic required.

Subscriber information (IMSI, etc.) is available on Gn side captures inside of the Create Session Request and similar GTP packets. From the Gn side, the payload is encapsulated in the GTP header and the source UE IP address is a pre-NAT private address.

Traffic captured on the Gi interface is post-NAT without the original private UE source IP address. No subscriber information is available on the Gi side traces.

Trace Logging

You can configure the MCC to log control plane information about a specific subscriber or subset of subscribers (for example, APN) up to a full debug 5 level.

The following command is an example of the required configuration:

```
an3000-mcm-slot7cpu1(config)# infrastructure logging event-
log policy TEST log-type debug filter-field
```

Possible completions:

```
apn component componentlogid cpu gname httpreqtype imei
imsi ipaddress macaddress msisdn nai npuflowid npunum
npuslot portnumber service severity slot teid vpn
```

```
infrastructure file-management profile eventlog IMSIDeb_
```

```
purge-thr                100
purge-low-thr            0
file-rotate-size-bytes   10485760
file-transfer-enabled    disabled
upload-frequency         1hour
upload-offset-hour       0
upload-offset-min        0
file-rotate-frequency    1hour
purge-file-older-than    30days
purge-enabled            disabled
compress-enabled         disabled
```

```
infrastructure logging event-log policy IMSIDeb
```

```
admin-state              enabled
log-type                 debug
filter-field             imsi
filter-value             <imsi>
                        exclude false
!
!
filter-field             severity
filter-value             Debug5
```

```

!
!
sinks                               IMSIDeb
type                                file
file-name                           IMSIDeb_
file-format                          ascii
!
!

```

```

an3000-mcm-slot7cpu1# infrastructure logging event-log
policy TEST log-type protocol protocol-logging-filter-field

```

Possible completions:

```

destination-ip-address destination-portnumber imei imsi ip-
address-pair ip-protocol ipaddress msisdn packet-direction
packet-type protocol source-ip-address source-portnumber

```

```

an3000-mcm-slot7cpu1# infrastructure logging event-log
policy TEST log-type protocol protocol-logging-filter-field
protocol filter-value

```

Possible completions:

```

dhcp diameter gtp gtpv0 gtpv1 gtpv2 radius

```

```

infrastructure logging event-log policy GTPCapture

```

```

admin-state enabled
log-type protocol
protocol-logging-filter-field protocol
filter-value gtpv1
filter-field interface
filter-value gn
exclude false
!
filter-value gp
exclude false
!
!
!
filter-value gtpv2
filter-field interface
filter-value s4
exclude false
!
filter-value s5

```

```

                                exclude false
                                !
                                filter-value s8
                                exclude false
                                !
                                filter-value s11
                                exclude false
                                !
                                !
                                !
                                !
                                !
                                sinks GTPCapture
                                type file
                                file-name GTPTrace_
                                file-format pcapng
                                !
                                !

```

In the example highlighted text, the logging configuration supports both debug and protocol type event logs. The logs are then written to a local or remote disk for analysis. The logs are stored in `/var/log/eventlog.lnk` and can be retrieved using SCP/SFTP or EMS.

The filename format is `<filenameprefix><YYYYMMDDHHMMSS>.gz` or `.pcap`. For example, `tos20140919165835.txt.gz` or `GTPTrace_20140922111150.pcapng`.

Trace Logging Considerations

The file management profile must be created before creating the event-log policy because every policy must reference a pre-existing profile. The profile does not need to be committed, but it should at least be entered into the pending configuration.

The log type can be either debug or protocol. If the debug log type is used, all traffic can be captured in binary or ASCII format and contains all flow information for the specified subscribers given the parameters. For efficiency, binary format is recommended, and you can use the EMS log viewer to decode the log.

If the protocol log type is used, all traffic can be captured in ASCII or pcapng format. Pcapng format is readable with Wireshark for analysis.

Setting up event logs is a valuable troubleshooting tool for capturing specific control and/or data plane traffic that transits MCC, as well as system logs.

Note: The information collected is subject to the configured criteria, and only control plane traffic is captured, not data plane. Also, while some logs can be configured for general use, using debug logs can affect system resources, so they should only be used for troubleshooting and then disabled.

DPI Statistics

Use the following command to display a overview of Deep Packet Inspection (DPI) statistics:

```
an3000-mcm-slot7cpu1# show zone default deep-packet-
inspection statistics grid-level

deep-packet-inspection statistics grid-level general 1
protocol-undetected-flows                0
detection-flows                          0
detection-packets                        0
detection-bytes                          0
decode-flows                             0
decode-packets                           0
decode-bytes                             0
protocol-undetected-packets              0
protocol-undetected-bytes                0
detection-total-packets                  0
detection-total-bytes                    0
decode-total-packets                     0
decode-total-bytes                       0
no-dpi-flows                             2466
no-dpi-total-packets                     222185
no-dpi-total-bytes                       189705273
detection-uplink-packets                 0
detection-uplink-bytes                   0
detection-downlink-packets               0
detection-downlink-bytes                 0
decode-uplink-packets                    0
decode-uplink-bytes                      0
decode-downlink-packets                  0
decode-downlink-bytes                    0
protocol-undetected-uplink-packets       0
protocol-undetected-uplink-bytes         0
protocol-undetected-downlink-packets     0
protocol-undetected-downlink-bytes       0
detection-total-uplink-packets           0
detection-total-uplink-bytes             0
detection-total-downlink-packets         0
detection-total-downlink-bytes           0
decode-total-uplink-packets              0
decode-total-uplink-bytes                 0
decode-total-downlink-packets            0
decode-total-downlink-bytes              0
no-dpi-total-uplink-packets              77233
no-dpi-total-uplink-bytes                 7799143
no-dpi-total-downlink-packets            144952
no-dpi-total-downlink-bytes              181906130
```

```

amqp-flows 0
amqp-uplink-packets 0
amqp-uplink-bytes 0
amqp-downlink-packets 0
amqp-downlink-bytes 0
application-undetected-flows 0
application-undetected-packets 0
application-undetected-bytes 0
application-undetected-uplink-packets 0
application-undetected-uplink-bytes 0
application-undetected-downlink-packets 0
application-undetected-downlink-bytes 0
deep-packet-inspection statistics grid-level tethering 1
  ConnectedDevice-flows 0
  ConnectedDevice-uplink-packets 0
  ConnectedDevice-uplink-bytes 0
  ConnectedDevice-downlink-packets 0
  ConnectedDevice-downlink-bytes 0
  TetheredDevice-flows 0
  TetheredDevice-uplink-packets 0
  TetheredDevice-uplink-bytes 0
  TetheredDevice-downlink-packets 0
  TetheredDevice-downlink-bytes 0
  TetheringEntitled-flows 0
  TetheringEntitled-uplink-packets-passed 0
  TetheringEntitled-uplink-bytes-passed 0
  TetheringEntitled-downlink-packets-passed 0
  TetheringEntitled-downlink-bytes-passed 0
  TetheredFlowsDetected-ttl 0
  TetheredFlowsDetected-user-agent 0
  TetheredFlowsDetected-TcpIpFingerprinting 0
  TetheredFlowsDetected-Gquic-user-agent 0
  TetheredFlowsReported-ttl 0
  TetheredFlowsReported-user-agent 0
  TetheredFlowsReported-TcpIpFingerprinting 0
  TetheredFlowsReported-Gquic-user-agent 0
deep-packet-inspection statistics grid-level web
HttpTransactions 1
  server-get_responses 0
  server-get-service-time-avg 0
  server-get-service-time-moving-avg 0
  proxy-get-responses 0
  proxy-service-time-avg 0
  proxy-service-time-moving-avg 0

```

Use the following parameters to display DPI statistics for a specific category:

```
business
conference
database
file-transfer
gaming
instant-messaging
mail
mobile
network-management
peer-to-peer
remote-control
social networking
standard
streaming
tunnel
voice-over-ip
web
```

You can specify specific applications within these categories. For example:

```
an3000-mcm-slot7cpu1# show zone default deep-packet-
inspection statistics grid-level applications 1 FaceTime-
downlink-bytes
```

BGP Routing

The **SUPPORTZONE** and **SAEGW** network contexts use BGP as the primary routing method and form BGP relationships with adjacent CE/PE routers.

The **bgp router summary** command displays the health of the BGP peer relationships:

```
an3000-mcm-slot7cpu1# show ip-protocols bgp router <network-
context> summary

bgp router GN

summary "Slot-Id: 2"

BGP router identifier 67.1.1.7, local AS number 67

BGP table version is 390

17 BGP AS-PATH entries

0 BGP community entries

Neighbor      V    AS MsgRcvd MsgSent   TblVer  InQ  OutQ Up/Down
State/PfxRcd  SSM/ISM Node
-----
```

```

160.0.0.1      4 65025      2411      2235      389      0      0 06:12:14
534           5

162.0.0.1      4 65025      2413      2270      390      0      0 06:12:14
534           6

4067:2266:aaaa:1::1 4 65025      2389      2235      380      0      0 06:12:14
103           6

4067:2299:aaaa:1::1 4 65025      2386      2244      380      0      0 06:12:14
103           5

```

an3000-mcm-slot7cpu1# **show ip-protocols bgp router <network-context> ipv4-ucast-af**

ipv4-ucast-af

route-table entry

BGP table version is 5736, local router ID is 107.77.163.1

Status codes: s suppressed, d damped, h history, * valid, > best, i - internal, l - labeled, m - multipath

S Stale

Origin codes: i - IGP, e - EGP, ? - incomplete

	Network	Next Hop	Metric	LocPrf	Weight	Path
* i	m0.0.0.0/0	172.20.0.37	0			0 65101 20057
* i	m	172.20.0.45	0			0 65101 20057
*> i	m	172.20.0.33	0			0 65101 20057
* i	m	172.20.0.41	0			0 65101 20057
*>	107.77.163.1/32	127.0.0.2	1		32768	?
*>	107.77.163.2/32	127.0.0.2	1		32768	?
*>	107.77.163.3/32	127.0.0.2	1		32768	?
*>	107.77.163.4/32	127.0.0.2	1		32768	?
*>	107.77.163.5/32	127.0.0.2	1		32768	?

Total number of prefixes 6

an3000-mcm-slot7cpu1# **show ip-protocols bgp router <network-context> ipv6-ucast-af**

```
ipv6-ucast-af
  route-table entry
BGP table version is 5737, local router ID is 107.77.163.1
Status codes: s suppressed, d damped, h history, * valid, > best, i -
internal, l - labeled, m - multipath
          S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop          Metric LocPrf Weight Path
*   m::/0            2600:300:2000:3407::2 (fe80::6664:9b0d:49b5:29c3)
                                   0              0 65101 20057 i
*   m                2600:300:2000:3407::6 (fe80::6664:9b0d:4bb5:29c7)
                                   0              0 65101 20057 i
*>  m                2600:300:2000:3407:: (fe80::6664:9b0d:48bc:2ac1)
                                   0              0 65101 20057 i
*   m                2600:300:2000:3407::4 (fe80::6664:9b0d:4abc:2ac5)
                                   0              0 65101 20057 i

*>  2600:387:0:900::/126
                                   ::              32768 ?
*>  2600:387:0:900::1/128
                                   ::1             1          32768 ?
*>  2600:387:0:900::2/128
                                   ::1             1          32768 ?
*>  2600:387:0:900::3/128
                                   ::1             1          32768 ?
*>  2600:387:0:900::4/126
                                   ::              32768 ?
*>  2600:387:0:900::4/128
                                   ::1             1          32768 ?
*>  2600:387:0:900::5/128
                                   ::1             1          32768 ?
```

Bidirectional Forwarding Detection

Bidirectional forwarding detection (BFD) allows for fast failure detection between the routing engine of BFD peered neighbors over multiple routing protocols (including static) with a minimum of overhead. BFD also improves failure detection time. However, it can also become a

point of failure if there are configuration or software issues related to the BFD protocol.

show ip-protocols bfd

Use the **interfaces** parameter for the BFD command to display the status of each BFD interface based on the responses to echo packets sent at configured intervals over that interface.

```
an3000-mcm-slot7cpu1# show ip-protocols bfd instance
<network-context> interfaces

interfaces entry

Interface: XXXX_1_1_5  ifindex: 68 state:    UP
Interface level configuration: NO ECHO, NO SLOW TMR
Timers in Milliseconds
Min Tx: 500  Min Rx: 500  Multiplier: 3

Interface: XXXX_2_2_5  ifindex: 69 state:    UP
Interface level configuration: NO ECHO, NO SLOW TMR
Timers in Milliseconds
Min Tx: 500  Min Rx: 500  Multiplier: 3

Interface: XXXX_3_1_5  ifindex: 70 state:    UP
Interface level configuration: NO ECHO, NO SLOW TMR
Timers in Milliseconds
Min Tx: 500  Min Rx: 500  Multiplier: 3

Interface: XXXX_4_2_5  ifindex: 71 state:    UP
Interface level configuration: NO ECHO, NO SLOW TMR
Timers in Milliseconds
Min Tx: 500  Min Rx: 500  Multiplier: 3
```

Use the **sessions** parameter of the BFD command to display the status of the peering between the BFD neighbors and indicators of stability, along with the configured/default timers.

```
an3000-mcm-slot7cpu1# show ip-protocols bfd instance
<network-context> sessions

sessions entry

Session Interface Index : 70      Session Index : 1
Lower Layer : IPv4                Version : 1
Session Type : Single Hop        Session State : Up
Local Discriminator : 1           Local Address : 172.20.0.42/32
```

```
Remote Discriminator : 1469      Remote Address :
172.20.0.41/32

Local Port : 49152              Remote Port : 3784

Options :

Diagnostics : None

Timers in Milliseconds :

Min Tx: 500                    Min Rx: 500
Multiplier: 3

Neg Tx: 2000                  Neg Rx: 2000          Neg
detect mult: 3

Min echo Tx: 200              Min echo Rx: 100        Neg echo
intrvl: 0

Storage type : 2

Sess down time : 00:00:00
Sess discontinue time : 00:00:00

Bfd GTSM Disabled

Bfd Authentication Disabled

UP Count : 0                  UPTIME : 7d20h53m

Protocol Client Info:

BGP-> Client ID: 7           Flags: 4

-----
```

HTTP Transaction Data Records (HTDR)

For each transaction processed by the MCC multi-protocol proxy, an HTTP Transaction Data Record (HTDR) entry is created at transaction termination. This may be on flow termination or when a new HTTP GET transaction is allocated for a current flow (such as a new YouTube video chunk as part of the full stream). When MCC is sized correctly, there is no additional impact on system resources.

HTDRs are stored locally on the system at */public/logs/datarecord.lnk*. The filename format is *<filename-prefix><YYYYMMDDHHMMSS>_<123>.csv*, for example, HTDR20170919103140_123.csv. They are stored natively as CSV files and a data-template parser can be built.

The following example shows sample output for a single entry:

```
'1411147899,221,354','111113076999899,999930769998990,3076999899,214
8630528,226,10,5,6,1,2606ae00b8001b000000000000000001,,reseller','MV
NO1,0a000003,1024,4a7de048,80,0','2,http%3a%2f%2fwww.google.com%2f%3
fgws_rd%3dssl,,0,0,Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.1;
.NET CLR
1.1.4322),,,203,0,203,0,0,1,0,0,0,9ba57302,6279,4a7de048,80,0','200,
text/html; charset=UTF-8;text/html; charset=UTF-
8,11514,11514,815,11514,815,11514,0,0,0',,,,,,
```

Time Information [HTDR: 1] Decoded below

Transaction Timestamp [sf : 1] 1411147899 (19-Sep-2014
17:31:39 GMT)

Request Latency (ms) [sf : 2] 221

Response Latency (ms) [sf : 3] 354

Subscriber [HTDR: 2] Decoded below

IMSI [sf : 1] 111113076999899

IMEI [sf : 2] 999930769998990

MSISDN [sf : 3] 3076999899

SessionID [sf : 4] 2148630528

Serving NW MCC [sf : 5] 226

Serving NW MNC [sf : 6] 10

Home PLMNID Length [sf : 7] 5

RAT Type [sf : 8] 6 (EUTRAN)

Roaming Status [sf : 9] 1 (True)

SGW/SGSN IP Address [sf :10]
2606ae00b8001b000000000000000001 (2606:ae00:b800:1b00::1)

APN (Mapped) [sf :11]

APN (Received) [sf :12] reseller

Client Connection [HTDR: 3] Decoded below

Gi Network Context [sf : 1] MVNO1

Source IP Address [sf : 2] 0a000003 (10.0.0.3)

Source Port [sf : 3] 1024

Destination IP Address [sf : 4] 4a7de048 (74.125.224.72)

Destination Port [sf : 5] 80

IP-Transparency [sf : 6] 0 (Disabled)

```
HTTP Request          [HTDR:  4] Decoded below

  HTTP Request Method      [sf : 1] 2 (GET)

  Client Request URL (Orig) [sf : 2]
http%3a%2f%2fwww.google.com%2f%3fgws_rd%3dssl

  Server Request URL (Real) [sf : 3]

  Client Protocol          [sf : 4] 0 (TCP)

  Server Protocol          [sf : 5] 0 (TCP)

  User-Agent               [sf : 6] Mozilla/4.0 (compatible;
MSIE 6.0; Windows NT 5.1; .NET CLR 1.1.4322)

  Content-Type             [sf : 7]

  Content-Length           [sf : 8]

  Client Request Header Size [sf : 9] 203

  Client Request Body Size  [sf :10] 0

  Server Request Header Size [sf :11] 203

  Server Request Body Size  [sf :12] 0

  Range Request            [sf :13] 0 (False)

  Request Successful        [sf :14] 1 (True)

  Chunked Request           [sf :15] 0 (False)

  Explicit Proxy Request    [sf :16] 0 (False)

  DNS Resolution Applied    [sf :17] 0 (False)

  Proxy local IP Address    [sf :18] 9ba57302 (155.165.115.2)

  Proxy local IP port       [sf :19] 6279

  Origin Server IP          [sf :20] 4a7de048 (74.125.224.72)

  Origin Server Port        [sf :21] 80

  Transaction Utilized BEF  [sf :22] 0 (Disabled)

HTTP Response          [HTDR:  5] Decoded below

  HTTP Response Status      [sf : 1] 200

  Server Content-Type        [sf : 2] text/html; charset=UTF-8

  Client Content-Type        [sf : 3] text/html; charset=UTF-8

  Server Content-Length      [sf : 4] 11,514

  Client Content-Length      [sf : 5] 11,514
```

```

Server Response Header Size    [sf : 6] 815
Server Response Body Size     [sf : 7] 11,514
Client Response Header Size   [sf : 8] 815
Client Response Body Size     [sf : 9] 11,514
Chunked Response               [sf :10] 0 (False)
Served From Cache              [sf :11] 0 (False)
Response Cached                [sf :12] 0 (False)

Compression                    [HTDR: 6] Field empty
Web Image Optimization         [HTDR: 7] Field empty
Video Optimization             [HTDR: 8] Field empty
Minification                   [HTDR: 9] Field empty
DNS Preemption                 [HTDR: 10] Field empty

```

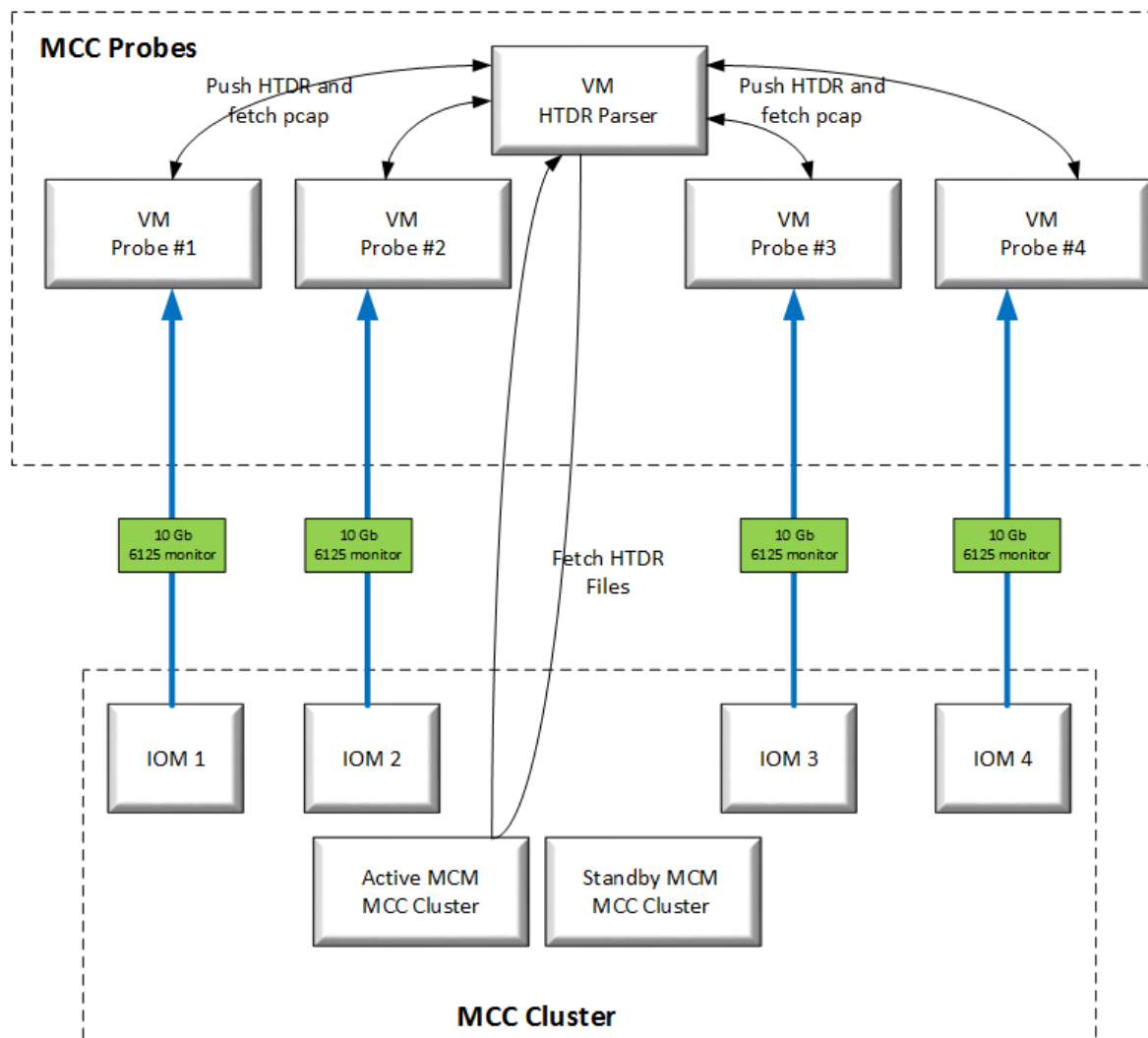
In the example highlighted text, HTDRs show subscriber information such as IMSI, IMEI, and MSISDN, the original private pre-NAT UE IP source address and port, and the public post-NAT UE IP source address and port.

HTDR Parser

HTTP Transaction Data Records (HTDR) can be used to parse through SGi pcaps to correlate SGi to a specific IMSI. Since most SGi traffic is proxy traffic, there is a public IP address on the MCC. The HTDR parser was developed to generate a SGi pcap for a specified IMSI.

[VMs in the HTDR System](#) shows the five VMs in the HTDR system. Four VMs are used for probes and one VM handles the HTDR files and correlation with the pcaps from the probes.

Figure 1-1 VMs in the HTDR System



HTDR Files

- All script files are stored in `/home/root`.
- All HTDRs and pcaps are stored in `/mnt/ramdisk`.

The following files are stored in the `/home/root` directory:

- **htdr-parser.sh** – This is the main script that starts up during bootup. A crontab runs this script when the VM is booted. The scripts builds the ramdisk, starts ntp timing, and changes the hostname for the VM.
- **htdrfind.pl** – This is the main perl script that runs when the user wants to generate an SGI pcap. This calls the following scripts:
 - **htdrlist.sh** – Provides the directory output of the `/mnt/ramdisk/htdr/htdrfiles` directory. It lists all current HTDR files.

- **htdrget.sh** – Clears out all old HTDRs in the `/mnt/ramdisk/htdr/htdrfiles/`. This calls the following script:
- **getfile.sh** – Greps all current HTDRs on the MCC.
- **htdrunzip.sh** – Provides the directory output of the `/mnt/ramdisk/htdr/unzip` directory. It lists all current HTDR files that are unzipped.
- **puthtdrx.sh** – Sends the parsed HTDR file to each probe using SFTP.

Probe Files

The four probe VMs also have scripts in the `/home/root` directory:

- **sdprobex.sh** – This is the main script that starts up during bootup. It starts the tshark that generates the pcaps. This script also creates the ramdisk and starts the ntp process. The following scripts are run from this script:
 - **htdrdirlist.sh** and **dirlistperl.pl** – Monitors the `/mnt/ramdisk/zhtdr` directory to see if any new HTDR files have been sent to the probes. If a new HTDR file is seen, it parses the file with the following script:
 - **htdr_parser.pl** – Parses the HTDR and creates the pcap in the `/mnt/ramdisk/zparsed-pcap/` directory.

Running the HTDR Parser

To run the HTDR parser, log into the Unix shell of the htldr-parser VM using Putty or any SSH client. The SSH port is 2222.

1. Use the following command to run the script:

```
root@HTDR-PARSER:~# perl /home/root/htdrfinder.pl

perl: warning: Setting locale failed.

perl: warning: Please check that your locale settings:

    LANGUAGE = "en_US:en",
    LC_ALL = (unset),
    LANG = "en_US.UTF-8"

are supported and installed on your system.

perl: warning: Falling back to the standard locale ("C").

Enter IMSI ----->
```

2. Enter the appropriate IMSI at the prompt.

If the IMSI is not present in any of the HTDRs, the following message is displayed:

```
sftp> IMSI 1234 is not present on any HTDR in the cluster
```

If the IMSI is present, the following output is displayed:

```
sftp>
```

```
Files pushed waiting for output pcaps  
my imsi is 111118352600990  
root@HTDR-PARSER:~#
```

HTDR Output

The HTDR output may take between 1 and 20 minutes to parse, depending on the size of the SGi data. When a pcap is parsed, it is saved in the following directory on the HTDR VM:
/mnt/ramdisk/zparsed-pcap/

When all four probes have completed their parsing, the four following files should be present:

/mnt/ramdisk/raw-pcap/output1.pcap

/mnt/ramdisk/raw-pcap/output2.pcap

/mnt/ramdisk/raw-pcap/output3.pcap

/mnt/ramdisk/raw-pcap/output4.pcap

Run the following script on the HTDR VM to merge the four files into one file:

```
root@HTDR-PARSER:~# sh /home/root/mergefile.sh
```

The script creates the following merged pcap:

/mnt/ramdisk/parsed-pcap/merged_SGi.pcap

Miscellaneous CLI Commands

alarms active

Use the **show alarms active** command to display current alarm details.

```
an3000-mcm-slot8cpu1# show alarms active

alarms active 265
trapName anLicenseCapacityEnforcement
xpath    /cluster[39]/MCC
snmpsev   critical
dateTime  "2020-11-30 15:29:3.220 +0000"
details   "Approaching or crossed licensed service capacity enforcement level. Upon
reaching X percent of the capacity enforcement level, a major event will be generated.
The event will become critical as the enforcement level is crossed."
```

See the *Affirmed Networks SNMP MIB and Alarm Reference Guide* for information about alarms supported on the system.

clear-config

Use the **clear-config** command to remove all the configuration from the MCC.

```
an3000-mcm-slot7cpu1# clear-config
```

Caution: You must **never** run this command under any circumstances in a production environment.

commit

Use the **commit** command (in Config mode) to execute the current set of changes to the MCC.

```
an3000-mcm-slot7cpu1(config)# commit
```

configuration commit list

Use the **show configuration commit list** command to display a list of the most recent changes made to the MCC.

```
an3000-mcm-slot7cpu1(config)# show configuration commit list

2020-02-27 11:08:01
SNo. ID      User      Client      Time Stamp      Label
Comment
~~~~ ~~~~   ~~~~~~   ~~~~~~   ~~~~~~   ~~~~~~
~~~~~
```

```
0    10967    admin    cli        2020-02-27 09:48:59
1    10966    admin    cli        2020-02-27 09:48:59
2    10965    admin    cli        2020-02-27 09:48:58
3    10964    admin    cli        2020-02-27 09:48:58
```

revert

Use the **revert** command (in Config mode) to abandon all uncommitted changes in the MCC.

```
an3000-mcm-slot7cpu1(config)# revert
```

set-prompt

Use the **set-prompt** command to customize the CLI prompt.

The following special options are supported:

***Note:** The `\` character is only supported for the options listed below.*

```
an3000-mcm-slot7cpu1# set-prompt

special options for prompts:

\d the date in "YYYY-MM-DD" format (e.g., "1978-06-10")
\h the hostname up to the first '.'
\H the hostname
\t the current time in 24-hour HH:MM:SS format
\T the current time in 12-hour HH:MM:SS format
\@ the current time in 12-hour am/pm format
\A the current time in 24-hour HH:MM format
\u the username of the current user
```

top

Use the **top** command to return to the top level of the CLI.

```
an3000-mcm-slot7cpu1# top
```

who

Use the **who** command to display currently logged on users. An asterisk appears next to the user who executed the command.

```
an3000-mcm-slot7cpu1# who
```

Session	User	Context	From	Proto	Date	Mode
66745	affirmed	snmp	10.32.3.149	udp	09:42:52	operational
*66713	admin	cli	169.254.147.81	ssh	09:35:02	operational

Session Management and Debugging

This section describes the recommended troubleshooting for problems related to EPS bearer activation and bearer traffic on the GGSN/PGW.

When you troubleshoot any issue, including Session Management, Affirmed Networks recommends that you check the active alarms and the cluster status using the following commands:

```
an3000-mcm-slot7cpu1# show cluster summary
an3000-mcm-slot7cpu1# show alarms active
an3000-mcm-slot7cpu1# show alarms history
an3000-mcm-slot7cpu1# show cluster node fault-summary
```

PDN Session Establishment Issues

Bearer activation issues occur for the following reasons:

- Signaling messages are not reaching the GGSN.
- Signaling messages are explicitly being rejected by the SGW or PGW.
- Signaling messages are accepted and the session is open but no data flows are created for the session.

Detecting Interface Connectivity Issues

Use the following command to determine whether session requests are being received by the MCC:

```
an3000-mcm-slot7cpu1# show zone default gateway debug-stats
session-setup pdngw service-level | select num-session-
setup-attempts
```

If session requests are being received it is assumed that the num-session-setup-attempts should be increasing when the command is run at 10-second intervals. If the session requests value is increasing, proceed to [step 1](#) , otherwise go to [Troubleshooting Gateway Signaling Rejects](#).

1. If requests are not being received, check network connectivity between the MCC and the adjacent GSN node (SGSN for 2G and 3G, MME for LTE) over the relevant interface. The ping test should always originate from the MCC's local endpoint. The following example shows commands for LTE troubleshooting:

```
an3000-mcm-slot7cpu1# ping <networkContext> -I <GTP_C_IP_Address> <GSN_Peer_IP_address>
```

LTE example:

```
an3000-mcm-slot7cpu1# ping [networkContext] [MME IP] [-I MCC-s11_loopback]
an3000-mcm-slot7cpu1# ping GnGp-NC 172.25.17.26 -I 107.77.171.1
PING 172.25.17.26 (172.25.17.26) from 107.77.171.1 : 56(84) bytes of data.
64 bytes from 172.25.17.26: icmp_req=1 ttl=57 time=0.502 ms
64 bytes from 172.25.17.26: icmp_req=2 ttl=57 time=0.476 ms
```

If the ping test from the loopback to the MME is successful, the problem is likely to exist between the MME and UE. If the ping fails, proceed to [step 2](#).

2. If the ping fails, test connectivity to the gateway. For example:

```
an3000-mcm-slot7cpu1# ping GnGp-NC 172.20.128.41 -I 107.77.171.1
PING 172.20.128.41 (172.20.128.41) from 107.77.171.1 : 56(84) bytes of data.
64 bytes from 172.20.128.41: icmp_req=1 ttl=64 time=0.455 ms
64 bytes from 172.20.128.41: icmp_req=2 ttl=64 time=0.424 ms
64 bytes from 172.20.128.41: icmp_req=3 ttl=64 time=0.447 ms
64 bytes from 172.20.128.41: icmp_req=4 ttl=64 time=0.441 ms
```

If the ping to the gateway from the loopback fails, a network connectivity issue exists between the MCC and the network.

If no connectivity issues are identified at this stage, verify the proper operation of the elements between the network and UE inclusive.

Troubleshooting Gateway Signaling Rejects

The MCC hosts GGSN/PGW functions, and in the absence of networking issues, see [Detecting Interface Connectivity Issues](#), use these functions to investigate session establishment issues.

PGW Signaling Rejects

To detect signaling rejects related to the PGW, first determine the number of active PGW sessions/bearers.

Use the following command and check the value for the following parameters:

```
an3000-mcm-slot7cpu1# show zone default gateway statistics
cluster-level call-performance pgw
```

```
current-sessions
```

active-sessions	Current number of active sessions
current-dual-stack-pdn-sessions	Current number of Dual Stack PDN Sessions
current-ipv4-pdn-sessions	Current number of IPv4 PDN Sessions
current-ipv6-pdn-sessions	Current number of IPv6 PDN Sessions
current-unique-subscribers	Current number of unique subscribers
bearers	Current number of bearers
dedicated-bearers	Current number of dedicated bearers
default-bearers	Current number of default bearers

Use the following command to determine which signaling procedure to the PGW failed:

```
an3000-mcm-slot7cpu1# show zone default gateway statistics
cluster-level call-performance pgw | include rejects
```

```
create-session-rejects
create-session-rejects-malformed-requests
create-session-rejects-unknown-apn
create-session-rejects-resource-unavailable
create-session-rejects-service-not-supported
create-session-rejects-system-failure
create-session-rejects-misc
delete-session-rejects
ue-initiated-dedicated-bearer-activation-rejects
pgw-initiated-dedicated-bearer-activation-rejects
mme-initiated-dedicated-bearer-deactivation-rejects
ue-initiated-dedicated-bearer-deactivation-rejects
pgw-initiated-dedicated-bearer-deactivation-rejects
modify-bearer-command-rejects
update-bearer-rejects
modify-bearer-rejects
create-pdp-context-rejects
create-pdp-context-rejects-malformed-requests
```

```
create-pdp-context-rejects-unknown-apn
create-pdp-context-rejects-resource-unavailable
create-pdp-context-rejects-service-not-supported
create-pdp-context-rejects-system-failure
create-pdp-context-rejects-misc
delete-pdp-context-rejects
ue-initiated-secondary-pdp-activation-rejects
ggsn-initiated-secondary-pdp-activation-rejects
sgsn-initiated-secondary-pdp-deactivation-rejects
ue-initiated-secondary-pdp-deactivation-rejects
ggsn-initiated-secondary-pdp-deactivation-rejects
sgsn-initiated-update-pdp-rejects
ggsn-initiated-update-pdp-rejects
change-notification-rejects
change-notification-rejects-malformed-requests
change-notification-rejects-unknown-apn
change-notification-rejects-resource-unavailable
change-notification-rejects-service-not-supported
change-notification-rejects-system-failure
change-notification-rejects-misc
num-suspend-notification-rejects
num-resume-notification-rejects
create-session-rejects
```

You can determine the exact cause code by checking the debug log or the Wireshark trace configured for the subscriber.

SGW Signaling Rejects

To detect signaling rejects related to the SGW, first determine the number of active subscribers.

Use the following command and check the value for the following parameters:

```
an3000-mcm-slot7cpu1# show zone default gateway statistics
cluster-level call-performance sgw
```

`active-pdn-connections` Current number of active PDN connections

`active-sessions` Current number of active sessions

To determine the number of requests rejected by the SGW, check the value for the following parameters:

`create-session-rejects-invalid-request`

`create-session-rejects-misc`

`create-session-rejects-system-failure`

`create-bearer-mme-rejects`

`create-bearer-rejects`

`create-bearer-rejects-invalid-requests`

`create-bearer-rejects-misc`

`create-bearer-rejects-system-failure`

`bearer-resource-fail-indication-invalid-requests`

`bearer-resource-fail-indication-misc`

`bearer-resource-fail-indication-system-failure`

You can determine the exact cause code by checking the debug log or the Wireshark trace configured for the subscriber.

PDN Session Payload Issues

When the bearer activation is successful but there is no payload or the expected payload values are not met, a payload handling issue exists.

The problem can be divided into two basic categories:

- Payload does not reach the GGSN/PGW
 - Payload is discarded by the GGSN/PGW
1. Check the uplink and downlink counters for packet drops.

Use the following commands to verify that uplink packets are reaching the gateway:

```
an3000-mcm-slot7cpu1# show zone default gateway statistics
cluster-level call-performance pgw GGSN | include bytes
```

```
an3000-mcm-slot7cpu1# show zone default gateway statistics
cluster-level call-performance pgw GGSN | include packets
```

Check the uplink counters:

```
uplink-bytes
```

uplink-packets

If the uplink counters are not increasing, it means that traffic is not being received by the gateway and you need to troubleshoot the network connectivity. To determine if downlink packets are being sent by the SAEGW, check the following counters:

downlink-bytes

downlink-packets

2. Use the following command to get more detailed information and check if the packets are being received and sent for a specific subscriber:

```
an3000-mcm-slot7cpu1# show subscriber pdn-session <id>
```

Check the following counters:

uplink-green-packets

uplink-green-bytes

downlink-green-packets

The same output gives detailed counters about the condition of dropped packets:

uplink-service-drop-packets

uplink-service-drop-bytes

uplink-no-tft-packets

uplink-no-route-packets

uplink-ttl-exhaust-packets

uplink-bad-header-packets

uplink-bad-ip-addr-packets

uplink-mtu-discard-packets

uplink-transmit-icmp-packets

uplink-send-error-packets

uplink-send-error-bytes

downlink-tcp-retransmission-packets

downlink-tcp-retransmission-bytes

downlink-diverted-returned-packets

downlink-error-drop-packets

downlink-error-drop-bytes

downlink-service-drop-packets

downlink-service-drop-bytes

downlink-no-tft-packets

```

downlink-no-route-packets
downlink-ttl-exhaust-packets
downlink-mtu-discard-packets
downlink-transmit-icmp-packets
downlink-bad-header-packets
downlink-send-error-packets
downlink-send-error-bytes
uplink-hairpin-policy-discard-packets
uplink-hairpin-error-discard-packets

```

Gx Issues

During session establishment and maintenance, the PCEF also communicates with the PCRF.

Use the following command to check for Diameter peering issues:

```

an3000-mcm-slot7cpu1# show service-construct interface aaa-
interface-group diameter peer-status

service-construct interface aaa-interface-group diameter
peer-status OCS-1 ocs2.sqa.affirmed.local
  realm                               sqa.affirmed.local
  ip-address                          10.100.241.2
  secondaryIpAddress                  ::
  dest-port                           3868
  host-port                           24641
  transport                           tcp
  role                                primary
  backup-peer-name                     ""
  status                               okay
  admin-status                         enabled
  last-state-change-timestamp          2020-01-22T15:21:55-0500
  origin-host                         ""
  origin-realm                        ""
  override-dest-hostname               ""

```

You should see the status of the peer as **okay**.

If the status is inactive, check the following output:

```

an3000-mcm-slot7cpu1# show service-construct interface aaa-
interface-group diameter peer-stats

```

```
service-construct interface aaa-interface-group diameter
peer-stats OCS-1 ocs2.sqa.affirmed.local
  realm                sqa.affirmed.local
  num-protocol-error    0
  num-permanent-failure 0
  num-routing-failure   0
  num-out-request       68524
  num-out-response      0
  num-in-request        0
  num-in-response       68524
```

Protocol errors occur at the base protocol level and may require per-hop attention. For example, a message routing error is a protocol error.

Common Diameter protocol errors are:

```
(3002) DIAMETER_UNABLE_TO_DELIVER (Initial/Terminate) The
diameter device cannot deliver the message to the desired
destination because the host described in the Destination-
Host AVP does not match a host within the associated realm.
```

```
(3004) DIAMETER_TOO_BUSY (Initial/Update/Terminate) The
diameter device cannot deliver the message to the desired
destination.
```

```
(3005) DIAMETER_LOOP_DETECTED (Initial/Terminate) A diameter
agent detected a loop while trying to get the message to the
desired destination.
```

Permanent failure errors (5xxx) inform the peer that the request failed and should not be re-attempted.

Use the following parameters to configure whether or not to send a terminate message upon receipt of a permanent failure error message:

```
an3000-mcm-slot7cpu1(config)# service-construct interface
pcrf-charging <pcrf-interface> pcrf-cca-error-send-terminate
true
```

Check the response counters to determine whether the PCRF server is replying to the PCEF. The PCRF server is replying to the messages when the responses match the requests. If the server is replying, check the Wireshark trace for a Diameter Success (Cause Code 2001) for the CEA.

If there is a Diameter success message from the PCEF, check the gateway logs. The log should contain the reason for not establishing the peer connection (e.g. CEA timeout, realm-name/host-name mismatch).

Use the following command to check the CEA timeout value and the hostname/realmname on the gateway side:

```
an3000-mcm-slot7cpu1# show running-config service-construct
interface aaa-interface-group diameter node-admin <pcrf-
interface-name>
```

Use the following command if the Diameter peer is established but you still observe a large number of CCR/CCA failures:

```
an3000-mcm-slot7cpu1# show service-construct interface aaa-
interface-group diameter statistics cluster-level peer
interface-type GX <pcrf-interface-name>
```

Gx Failure Handling

If the Diameter link between the PCEF and PCRF fails, you can configure the MCC to continue to set up the call using a locally configured policy.

Use the following commands to configure the call setup:

```
an3000-mcm-slot7cpu1(config)# service-construct interface
pcrf-charging PAS_INTF

an3000-mcm-slot7cpu1(config)# pcrf-down-default-gx-call-
setup enabled
```

Though service may be restored, any network functionality supported by the Gx interface may not be available (for example, charging, blocking for insufficient balance, throttling).

Default QoS Policy

If the Gx service fails on the MCC and the pcrf-down-default-gx-call-setup flag is set to enabled, the default behavior of the MCC is to assign the user with a default QoS policy. You can use counters that track the assignment of the default QoS policy to diagnose Gx issues. If the counters are increasing, assume there is an underlying issue.

Use the following command to display the relevant counters:

```
an3000-mcm-slot7cpu1# show zone default gateway statistics
cluster-level call-performance pgw | include default
```

default-bearers	37950
interval-peak-default-bearers	39385
number-of-sessions-setup-with-default-policy	735
number-of-active-sessions-with-default-policy	2
delete-pdp-context-drops-teardown-ind-missing-default-bearer	0

CALEA Issues

The Communications Assistance for Law Enforcement Act (CALEA) interface provides surveillance capabilities to assist law enforcement. For details, see the *Affirmed Networks Lawful Intercept Module Specification and Provisioning Guide*.

To execute any CALEA CLI commands, you must be logged in as the CALEA user:

- Username: calea
- Default Password: calea

show calea interface-status

Use the **show calea interface-status** command to display the status and utilization of the individual binds between the relevant CALEA interface and the remote server. To display the CALEA Interface Table status, you must be in Exec mode.

```
an3000-mcm-slot7cpu1# show calea interface-status

calea interface-status x2-ber-hi-allen

transport-type                tcp
interface-type                x2
interface-format-type         HI_format
operational-status            up
number-of-agent-instances     1
number-of-tpkt-keep-alive-messages-sent 11192
number-of-tpkt-keep-alive-ber-encode-fail-count 0
number-of-tpkt-keep-alive-xer-encode-fail-count 0

calea interface-status x2-ber-hi-bothell

transport-type                tcp
interface-type                x2
interface-format-type         HI_format
operational-status            up
number-of-agent-instances     1
number-of-tpkt-keep-alive-messages-sent 11203
number-of-tpkt-keep-alive-ber-encode-fail-count 0
number-of-tpkt-keep-alive-xer-encode-fail-count 0

calea interface-status x3-ber-hi-allen
```

```

transport-type                tcp
interface-type                x3
interface-format-type         HI_format
operational-status            up
number-of-agent-instances     1
number-of-tpkt-keep-alive-messages-sent 11172
number-of-tpkt-keep-alive-ber-encode-fail-count 0
number-of-tpkt-keep-alive-xer-encode-fail-count 0
calea interface-status x3-ber-hi-bothell
transport-type                tcp
interface-type                x3
interface-format-type         HI_format
operational-status            up
number-of-agent-instances     1
number-of-tpkt-keep-alive-messages-sent 11203
number-of-tpkt-keep-alive-ber-encode-fail-count 0
number-of-tpkt-keep-alive-xer-encode-fail-count 0

```

show calea interface-group-status

Use the **interface-group-status** parameter to display details on configuration and performance for the CALEA group defined on the MCC. To display the CALEA Interface Group status, you must be in Exec mode.

```

an3000-mcm-slot7cpu1# show calea interface-group-status
calea interface-group-status x2-ber-hi-allen
role                primary
interface-group-type x2
output-type         ber
interfaceFormat-type HI_format
operational-status  up
iri-buffer-size     1024000
iri-buffer-current-size 0
iri-buffer-num-packets 0

```

iri-buffer-high-water-mark	0
iri-buffer-num-buffer-size-exceeded	0
iri-buffer-num-dropped-packets	0
number-of-agent-instances	1
number-of-x2-events-offered	21
number-of-transmit-ok	21
number-of-transmit-fail-bad-interface-group	0
number-of-transmit-fail-interface-group-down	0
number-of-transmit-fail-no-space	0
number-of-x2-encode-fail-no-buffer	0
number-of-x2-ber-encode-fail	0
number-of-x2-xer-encode-fail	0
number-of-targets	0
total-cumulative-LI-Sessions	10
currently-active-li-calls	0
total-number-of-intercepted-packets-for-current-active-calls	0
number-of-x2-events-for-current-active-calls	0
total-number-of-all-terminated-LI-sessions	10
total-number-of-terminated-LI-sessions-due-to-call-disconnect	8
total-number-of-terminated-LI-sessions-due-to-deprovision	2
calea interface-group-status x2-ber-hi-bothell	
role	secondary
interface-group-type	x2
output-type	ber
interfaceFormat-type	HI_format
operational-status	up
number-of-agent-instances	1
number-of-transmit-ok	0
number-of-transmit-fail-bad-interface-group	0
number-of-transmit-fail-interface-group-down	0
number-of-transmit-fail-no-space	0
calea interface-group-status x3-ber-hi-allen	

role	primary
interface-group-type	x3
output-type	ber
interfaceFormat-type	HI_format
operational-status	up
cc-buffer-size	51200000
cc-buffer-current-size	0
cc-buffer-num-packets	0
cc-buffer-high-water-mark	0
cc-buffer-num-buffer-size-exceeded	0
cc-buffer-num-dropped-packets	0
number-of-agent-instances	1
number-of-x3-events-offered	48970
number-of-x3-event-encode-failures-no-buffer	0
number-of-x3-event-xer-encode-failures	0
number-of-x3-event-ber-encode-failures	0
number-of-transmit-ok	48970
number-of-transmit-fail-bad-interface-group	0
number-of-transmit-fail-interface-group-down	0
number-of-transmit-fail-no-space	0
number-of-x3-encode-fail-no-buffer	0
number-of-x3-ber-encode-fail	0
number-of-x3-xer-encode-fail	0
number-of-targets	0
total-cumulative-LI-Sessions	6
currently-active-li-calls	0
total-number-of-intercepted-packets-for-current-active-calls	0
number-of-x3-events-for-current-active-calls	0
calea interface-group-status x3-ber-hi-bothell	
role	secondary
interface-group-type	x3
output-type	ber

interfaceFormat-type	HI_format
operational-status	up
number-of-agent-instances	1
number-of-transmit-ok	0
number-of-transmit-fail-bad-interface-group	0
number-of-transmit-fail-interface-group-down	0
number-of-transmit-fail-no-space	0

show calea service-status

Use the **service-status** parameter to display a consolidated view of CALEA across all interfaces (X1, X2, X3). To display the CALEA service status, you must be in Exec mode.

```
an3000-mcm-slot7cpu1# show calea service-status
calea service-status all
operational-status                                up
number-of-agent-instances                          1
number-of-x1-1-interfaces                          0
number-of-x2-interfaces                            2
number-of-x3-interfaces                            2
number-of-bad-events-received                      0
number-of-rx-events-dropped-no-target              0
number-of-x2-events-received                       21
number-of-x3-events-received                       56821
number-of-transmit-ok                              48991
number-of-transmit-fail-bad-interface              0
number-of-transmit-fail-interface-down             0
number-of-transmit-fail-no-space                   0
number-of-echo-requests-received                   0
number-of-echo-requests-sent                       0
number-of-echo-responses-received                  0
number-of-echo-responses-sent                      0
number-of-echo-timeouts                            0
number-of-targets                                  0
total-cumulative-LI-Sessions                       10
```

currently-active-li-calls	0
currently-camp-on-triggers	0
number-of-x2-output-events	21
number-of-x3-output-events	48970
number-of-x2-output-events-failed	0
number-of-x2-events-dropped-no-interface	0
number-of-x2-encode-fail-no-buffer	0
number-of-x2-xer-encode-fail	0
number-of-x2-ber-encode-fail	0
number-of-x3-output-events-failed	0
number-of-x3-events-dropped-no-interface	0
number-of-x3-event-encode-failures-no-buffer	0
number-of-x3-xer-encode-fail	0
number-of-x3-ber-encode-fail	0
total-number-of-intercepted-packets-for-current-active-calls	0
number-of-x2-events-for-current-active-calls	0
number-of-x3-events-for-current-active-calls	0
total-number-of-all-terminated-LI-sessions	10
total-number-of-terminated-LI-sessions-due-to-call-disconnect	8
total-number-of-terminated-LI-sessions-due-to-deletion-of-context	0
total-number-of-terminated-LI-sessions-due-to-deprovision	2

GTPP Issues

GTP Prime (GTPP) is an IP-based protocol implemented over the Ga interface that can be used with both TCP and UDP for transporting charging data from the gateways (Charging Data Function (CDF)) to the Charging Gateway Function (CGF). The CDF function implemented in the MCC is bound to two CGFs in active/standby mode and the service is hosted in the SUPPORTZONE.

Before proceeding, check the active alarms and the cluster status as described in [Session Management and Debugging](#).

gtp node-status

Use the **show service-construct interface aaa-interface-group gtp node-status** command to display the operational status of the local GTPP instance:

```
an3000-mcm-slot7cpu1# show service-construct interface aaa-
interface-group gtp node-status
```

```
service-construct interface aaa-interface-group gtp node-status
```

							SEQUENCE	
			OPERATIONAL	HOST	HOST IP	DROPPED	DUPLICATE	OVER RUN
NAME	ADMINSTATUS	OPERSTATUS	REASON	PORT	ADDRESS	PACKETS	PACKETS	PACKETS

GA-XXXX	enabled	inservice	-	6101	107.77.163.9	0	0	0

gtp peer-status

Use the **gtp peer-status** parameter to check the GTPP peering state between the MCC and the GTPP endpoint:

```
an3000-mcm-slot7cpu1# show service-construct interface aaa-
interface-group gtp peer-status
```

```
service-construct interface aaa-interface-group gtp peer-
status GA-XXXX NDC2C_OCG7_1
```

```
ip-address          172.18.250.231
```

```
port                6101
```

```
transport           udp
```

```
role                primary
```

```
adminstatus         enabled
```

```
operstatus          okay
```

```
dataRecordRetryCount 3
```

```
dataRecordRespTimeout 5
```

```
echoRetryCount       3
```

```
echoRespTimer        10
```

```
periodicHealthCheckTimer 60
```

```
service-construct interface aaa-interface-group gtp peer-
status GA-XXXX NDC2C_OCG7_2
```

```
ip-address          172.18.250.236
```

```
port                6201
```

```
transport           udp
```

```
role                secondary
```

```
adminstatus         enabled
```

```
operstatus          okay
dataRecordRetryCount 3
dataRecordRespTimeout 5
echoRetryCount      3
echoRespTimer       10
periodicHealthCheckTimer 60
```

If the peer-status is enabled but the operational status is not `okay`, you need to verify connectivity between the MCC and the peer. Use the host IP address (see [gtpm node-status](#) for the host IP address) as the source address and send the ping from the SUPPORTZONE.

```
an3000-mcm-slot7cpu1# ping zzzz 172.18.250.231 -I
107.77.163.9

Warning: Permanently added '[169.254.100.9]:2020' (RSA) to the list of
known hosts.

PING 172.18.250.231 (172.18.250.231) from 107.77.163.9 : 56(84) bytes of
data.

64 bytes from 172.18.250.231: icmp_req=1 ttl=252 time=0.405 ms
64 bytes from 172.18.250.231: icmp_req=2 ttl=252 time=0.322 ms
64 bytes from 172.18.250.231: icmp_req=3 ttl=252 time=0.427 ms
64 bytes from 172.18.250.231: icmp_req=4 ttl=252 time=0.348 ms
64 bytes from 172.18.250.231: icmp_req=5 ttl=252 time=0.351 ms
```

gtpm peer-stats

Use the `gtpm peer-stats` parameter to display a consolidated view of the GTPM performance giving the number of aggregate messages and generic errors sent to and from a peer.

```
an3000-mcm-slot7cpu1# show service-construct interface aaa-
interface-group gtpm peer-stats

service-construct interface aaa-interface-group gtpm peer-
stats
```

		NUM	NUM	NUM						
NUM										
LINK		PROTOCOL	PERMANENT	ROUTING	NUM OUT	NUM OUT	NUM IN	NUM IN		
HOST NAME	NAME	ERROR	FAILURE	FAILURE	REQUEST	RESPONSE	REQUEST	RESPONSE		
BOUNCES										

GA-XXXX	YYYY_ZZZ_1	0	0	0	4229	73912	73912	3931	0	
GA-XXXX	YYYY_ZZZ_2	0	0	0	2463	73913	73913	2463	0	

gtp statistics

Use the **gtp statistics** parameter to display information about the status of communication between each GTP peer.

```
an3000-mcm-slot7cpu1# show service-construct interface aaa-
interface-group gtp statistics
```

```
service-construct interface aaa-interface-group gtp
statistics cluster-level peer node-name GA-XXXX YYYY_ZZZ_1
```

num-echo-req-tx	2462
num-echo-req-rx	73880
num-echo-resp-tx	73880
num-echo-resp-rx	2462
num-node-alive-req-tx	3
num-node-alive-req-rx	0
num-node-alive-resp-tx	0
num-node-alive-resp-rx	0
num-redirection-req-rx	0
num-redirection-resp-tx	0
num-data-record-transfer-req-tx	1763
num-data-record-transfer-resp-rx	1466
num-potential-duplicate-data-record-rx	0
num-potential-duplicate-empty-data-record-tx	0
num-cancel-request-tx	0
num-release-request-tx	0
num-req-fulfilled-rx	0
num-req-accepted-rx	0
Discontinuity Timestamp	1430802249

```
service-construct interface aaa-interface-group gtp
statistics cluster-level peer node-name GA-XXXX YYYY_ZZZ_2
```

num-echo-req-tx	2462
num-echo-req-rx	73882
num-echo-resp-tx	73882
num-echo-resp-rx	2462

num-node-alive-req-tx	3
num-node-alive-req-rx	0
num-node-alive-resp-tx	0
num-node-alive-resp-rx	0
num-redirection-req-rx	0
num-redirection-resp-tx	0
num-data-record-transfer-req-tx	0
num-data-record-transfer-resp-rx	0
num-potential-duplicate-data-record-rx	0
num-potential-duplicate-empty-data-record-tx	0
num-cancel-request-tx	0
num-release-request-tx	0
num-req-fulfilled-rx	0
num-req-accepted-rx	0
Discontinuity Timestamp	1430802249

GiGW Subscriber Learning Limitations

The following behavior is related to GiGW session learning from RADIUS Accounting-Request messages.

- For emergency sessions (IMSI==0) without MSISDN (Calling-Station-Id), GiGW maintains only one session, therefore, if another emergency call occurs, MCC replaces the first session with the second session.
- For fixed-line subscribers, the username takes precedence over MSISDN. That is, if a username exists, MCC uses it to anchor the pseudo IMSI. If no username exists, MSISDN is used to anchor the pseudo IMSI.
- For WAG subscribers, the username and Calling-Station-Id should be in MAC address format.

SLB Load Status Table

The Steering Load Balancer (SLB) load status table displays the load distribution status across the gateways.

```
an3000-mcm-slot7cpu1# show zone default gateway slb-load-status
```

gateway slb-load-status

GATEWAY			LOAD	CURRENT	ABSOLUTE
TYPE	NAME	SERVICEID	STATE	LOAD	VALUE

sgw	SGW_100_11_1_0_SGW_0	152	MinorLow	0	0
sgw	SGW_100_11_1_12_SGW_24	741	MinorLow	0	0
sgw	SGW_100_11_1_16_SGW_32	747	MinorLow	0	0
sgw	SGW_100_11_1_20_SGW_40	751	MinorLow	0	0
sgw	SGW_100_11_1_24_SGW_48	755	MinorLow	0	0
sgw	SGW_100_11_1_28_SGW_56	759	MinorLow	0	0
sgw	SGW_100_11_1_4_SGW_8	699	MinorLow	0	0
sgw	SGW_100_11_1_8_SGW_16	721	MinorLow	0	0
sgw	SGW_100_12_1_13_SGW_26	742	MinorLow	0	0
sgw	SGW_100_12_1_17_SGW_34	748	MinorLow	0	0
sgw	SGW_100_12_1_1_SGW_2	164	MinorLow	0	0
sgw	SGW_100_12_1_21_SGW_42	752	MinorLow	0	0
sgw	SGW_100_12_1_25_SGW_50	756	MinorLow	0	0
sgw	SGW_100_12_1_29_SGW_58	760	MinorLow	0	0
sgw	SGW_100_12_1_5_SGW_10	703	MinorLow	0	0
sgw	SGW_100_12_1_9_SGW_18	725	MinorLow	0	0
sgw	SGW_100_13_1_10_SGW_20	730	MinorLow	0	0
sgw	SGW_100_13_1_18_SGW_36	749	MinorLow	0	0
sgw	SGW_100_13_1_22_SGW_44	753	MinorLow	0	0
sgw	SGW_100_13_1_26_SGW_52	757	MinorLow	0	0
sgw	SGW_100_13_1_2_SGW_4	176	MinorLow	0	0
sgw	SGW_100_13_1_30_SGW_60	761	MinorLow	0	0
sgw	SGW_100_13_1_6_SGW_12	708	MinorLow	0	0
sgw	SGW_100_14_1_11_SGW_22	736	MinorLow	0	0
sgw	SGW_100_14_1_15_SGW_30	744	MinorLow	0	0
sgw	SGW_100_14_1_19_SGW_38	750	MinorLow	0	0
sgw	SGW_100_14_1_23_SGW_46	754	MinorLow	0	0
sgw	SGW_100_14_1_27_SGW_54	758	MinorLow	0	0
pgw	PDNGW_100_11_1_0_PDNGW_0	145	MinorLow	0	0
pgw	PDNGW_100_11_1_12_PDNGW_24	589	MinorLow	0	0
pgw	PDNGW_100_11_1_16_PDNGW_32	611	MinorLow	0	0
pgw	PDNGW_100_11_1_20_PDNGW_40	633	MinorLow	0	0
pgw	PDNGW_100_11_1_24_PDNGW_48	655	MinorLow	0	0
pgw	PDNGW_100_11_1_28_PDNGW_56	677	MinorLow	0	0
pgw	PDNGW_100_11_1_4_PDNGW_8	545	MinorLow	0	0
pgw	PDNGW_100_11_1_8_PDNGW_16	567	MinorLow	0	0
pgw	PDNGW_100_12_1_13_PDNGW_26	593	MinorLow	0	0
pgw	PDNGW_100_12_1_17_PDNGW_34	615	MinorLow	0	1
pgw	PDNGW_100_12_1_1_PDNGW_2	157	MinorLow	0	0
pgw	PDNGW_100_12_1_21_PDNGW_42	637	MinorLow	0	0
pgw	PDNGW_100_12_1_25_PDNGW_50	659	MinorLow	0	0
pgw	PDNGW_100_12_1_29_PDNGW_58	681	MinorLow	0	0
pgw	PDNGW_100_12_1_5_PDNGW_10	549	MinorLow	0	0

Accessing the Remote Console

Use the .NET Integrated Remote Console Screen (.NET IRC) to access the blade.

JAVA IRC and the Mobile App are also valid access methods, but these have more dependencies, so .NET IRC is used in this example.

Figure 2-1 Remote Console Screen

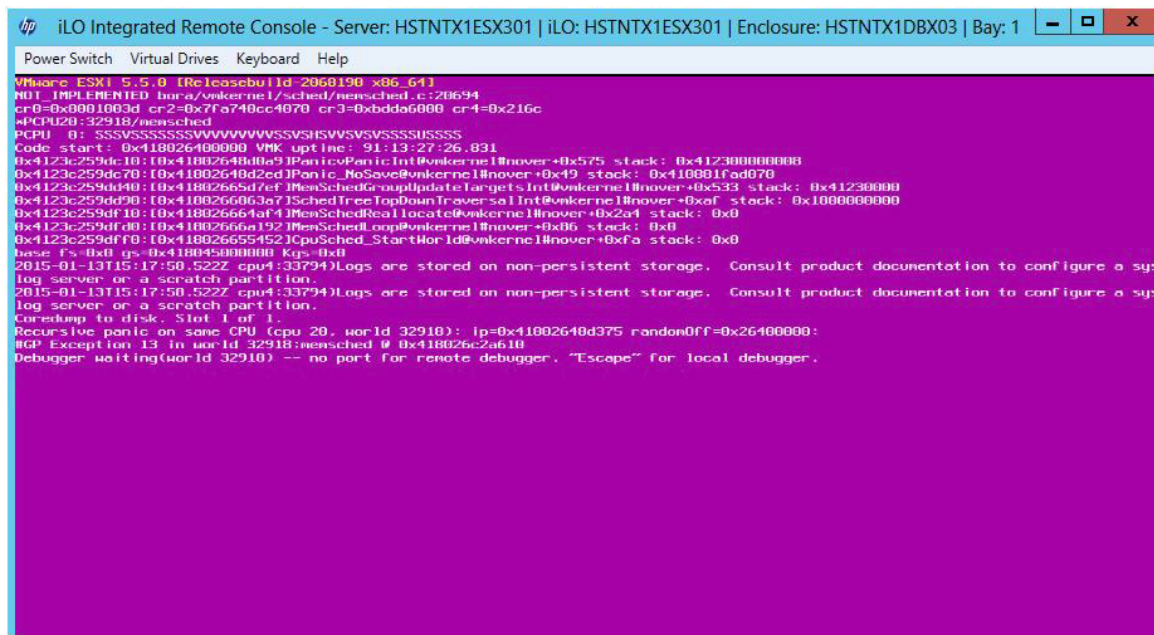
The screenshot displays the HP iLO 4 ProLiant DL380 Gen9 interface. The top navigation bar includes the HP logo, iLO 4 ProLiant DL380 Gen9, and user information: Local User: Administrator, iLO Hostname: ILOCZJ5290JRP, Home, and Sign Out. A left sidebar lists various management categories: Information (Overview, System Information, iLO Event Log, Integrated Management Log, Active Health System Log, Diagnostics, Location Discovery Services, Insight Agent), iLO Federation, Remote Console (Remote Console), Virtual Media, Power Management, Network, Remote Support, and Administration. The main content area is titled 'Remote Console - iLO Integrated Remote Console' and features tabs for Launch, Java, Hot Keys, and Security. The 'Launch' tab is active, showing three sections: .NET Integrated Remote Console (.NET IRC), Java Integrated Remote Console (Java IRC), and HP iLO Mobile App. The .NET IRC section includes a description, a note about Windows compatibility, and a '.NET Framework Detection' table. The Java IRC section includes a description and a 'Launch' button. The HP iLO Mobile App section includes a description, a download link, and a 'Learn More' button.

Compatible Versions	Status
3.5 (Full)	
4.0 (Full)	
4.5	OK

Recovering from a Fatal Error

When the VM kernel experiences a fatal error, the system displays a purple [Fatal Error Diagnostic Screen](#). This indicates the blade and all VMs hosted on the blade are down, as shown by the `show cluster summary` CLI command. All nodes hosted on the same blade will show the CPU state as down at the same time.

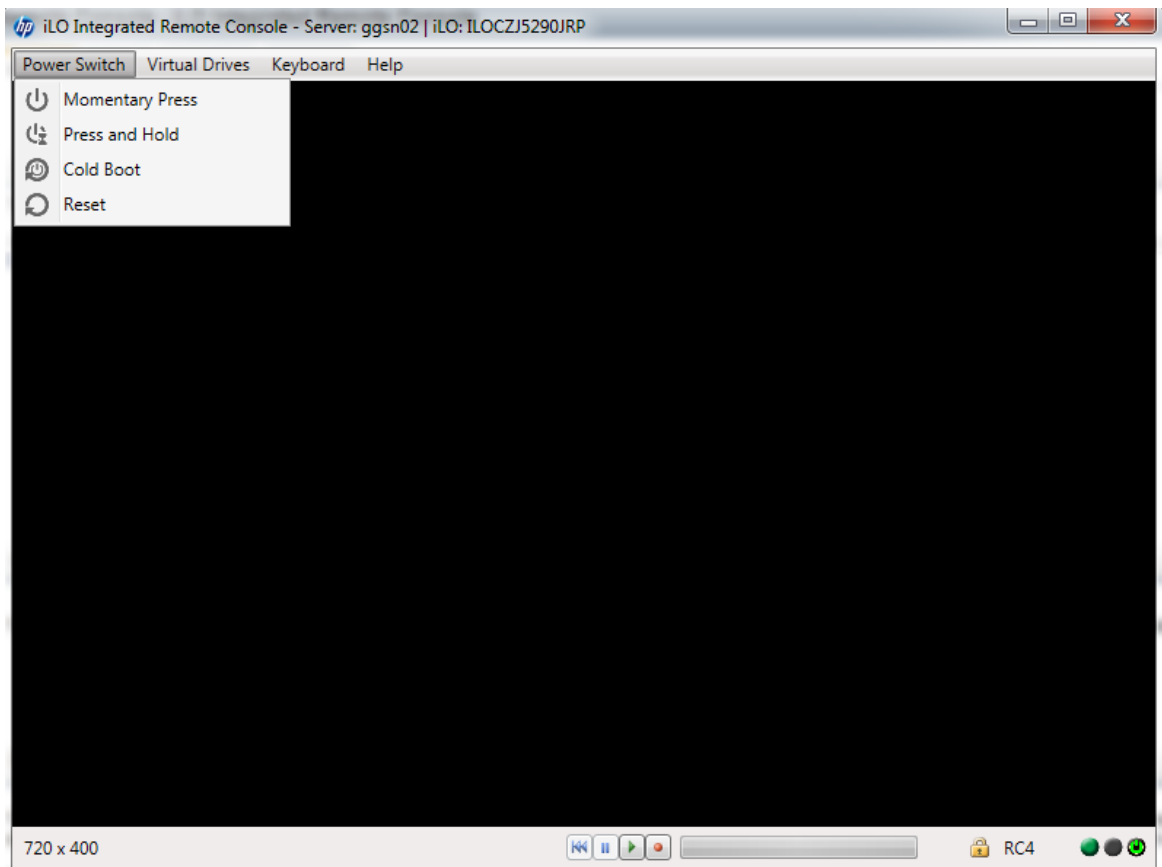
Figure 2-2 Fatal Error Diagnostic Screen



Before rebooting the blade, collect the fault files that are stored in the `/var/log/core` folder. Affirmed Networks Support uses these files to diagnose issues.

[Integrated Remote Console Power Switch](#) shows the power options. Reboot the blade using one of the reboot options from the **Power Switch** menu. [Remote Console Power Switch Options](#) describes the function of each option.

The blade normally takes approximately five minutes to reboot. Track the progress of the VMs using the `show cluster` CLI command.

Figure 2-3 Integrated Remote Console Power Switch**Table 2-1 Remote Console Power Switch Options**

Options	Description
Momentary Press	The same as pressing the physical power button. If a server is powered off, a momentary press will turn the server power on.
Press and Hold	The same as pressing the physical power button for five seconds and then releasing it.
Cold Boot	Immediately removes power from the server. Processors, memory, and I/O resources lose main power. The server will restart after approximately six seconds. This option circumvents the graceful shutdown features of the operating system.
Reset	Forces the server to warm-boot. CPUs and I/O resources are reset. This option circumvents the graceful shutdown features of the operating system.

Configuring Interfaces

This section describes useful information on configuring interfaces.

Configuring the SWu as an Access Interface

An access interface is any ip-interface that handles Encapsulated Security Payload (SWu interface to a UE) or L2oGRE (towards an access point) traffic. Any ip-interface handling that type of traffic *must* be configured as an access interface.

Use the following command to configure the SWu as an access interface:

```
an3000-mcm-slot7cpu1(config)# network-context <network-  
context-name> access-interface <ip-interface>
```

where

Parameter	Description
network-context	Specifies the name of the access network context.
access-interface	Specifies the IP Interface to be designated as an access interface.

Technical Support

This section describes:

- How to access the Affirmed Networks customer portal to open a support ticket
- What to include when creating a ticket
- Contacting Customer Support

Opening a Support Ticket

For non-critical cases, Affirmed Networks recommends that customers use the online ticket system as the primary option to report issues. The system allows Affirmed support to track a ticket during its lifetime (for SLA adherence), communicate with the customer, and escalate tickets to engineering. Access the customer portal at:

<https://system.netsuite.com/app/login/secure/privatelogin.nl?c=3505467>

Click the **Contact Support** link in the [Open a New Case in the Customer Portal](#) to open a ticket.

Figure 4-1 Open a New Case in the Customer Portal



[New Case Required Fields](#) shows the required fields, marked with a *.

Figure 4-2 New Case Required Fields

Completing the Ticket Information in the Customer Portal

Complete the following fields to create a ticket:

1. Incident Subject:

Enter a brief description of the issue.

2. Severity Level:

Select the severity level for the issue from the drop-down list: Critical, Major, Minor, or Info.

3. Product:

Select the product. For example, PGG, EMS, and vCenter.

4. Software Release Affected:

Enter the software release that is affected.

Use the **show image-management status** command to obtain the release number:

```
image-management status active 18 7 1
admin-version          12.0.0.0-1181.REL
operational-version    12.0.0.0-1181.REL
activation-status      Idle
rollback-version       12.0.0.0-1172.REL
prepared-version       NotAvailable
```

5. Location:

Enter the location of the site where the issue exists.

6. Issue Detail:

Describe the issue in as much detail as possible and include the following:

- A percentage estimate of the number of affected subscribers.

- Services that are affected.
- Information derived from troubleshooting.
- Events that may have led to the issue.

7. Technical Support Information Logs (when applicable):

Use the **save-tech-support-info** command to create detailed system logs that Affirmed Networks Support uses to diagnose issues. These logs can be very large; check that the file system has sufficient space before you generate the logs. The */public* folder should be below 80%.

```
an3000-mcm-slot7cpu1# save-tech-support-info
```

Use the following optional parameters to customize the logs:

```
--nofaults --nologs --noroutinginfo --routinginfoverbose --  
numdays=N --numhours=M --datarecords --psm
```

Depending on the MCC release you are running, you may see **--nocorefiles** or **--nomemimages** instead of **--nofaults**.

For EMS issues, the *emslogdump.pl* dump file is stored in the following folder:

```
/opt/Affirmed/NMS/bin/.emslogdump.pl
```

Attaching Files to Tickets

The following file types are supported as attachments to a ticket:

.bmp, .css, .csv, .doc, .dwg, .eml, .gif, .gx, .htm, .html, .ico, .img, .jpg, .jpeg, .js, .mps, .mpeg, .mpg, .mpp, .pdf, .pjpeg, .png, .ppl, .ps, .qt, .rtf, .sms, .swf, .tif, .tiff, .txt, .vsd, .xls, .xml, .zip.

If the file type you want to attach is not supported:

- Add the file to a .zip file; or
- Append .txt to the filename (for example, filename.log.txt) and attach the file to the ticket.

Customer Support

Affirmed Networks provides technical support 24 hours a day, 7 days a week (24 x 7) for any customer with an active maintenance contract.

To contact support:

E-mail: AN_support_operation@microsoft.com

24 x 7 Telephone support:

Local: +1-978-268-0800

Toll-free: +1-888-283-2838